International Association of Oil & Gas Producers

REPORT
2023pfh

JULY
2024

DATA SERIES

Safety performance indicators – Process safety events – 2023 data

Tier 1 PSE, fatal incident, and high potential event reports

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

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Safety performance indicators – Process safety events – 2023 data

Tier 1 PSE, fatal incident and high potential event reports

Revision history

VERSION	DATE	AMENDMENTS
1.0	July 2024	First release

Contents

TIER 1 PROCESS SAFETY EVENTS 2023

Tier 1 PSE are predominantly lagging indicators related to Loss of Primary Containment (LOPC) referred to as a Process Safety Event (PSE). The Tier 1 KPI records events with greater consequence within the four-tier approach.

For more information see the introduction and IOGP Report 456, Process Safety – Recommended Practice on Key Performance Indicators IOGP has been gathering Tier 1 PSE narrative reports from its Members since 2013. These include Tier 1 PSE reported both by companies and contractors. Event reports are categorized by region, country, location (onshore/offshore), cause, activity at the time of the event, and, from 2019 onwards, point of release.

The information provided here is not detailed; often the root cause of an incident cannot be established. However, the information should assist organizations to identify likely hazards, human particular, it allows organizations to question whether their own Safety Management System would have prevented the event occurring and mitigated its consequences.

The database from 2020 onwards is available and searchable at https://data.iogp.org/ProcessSafety/Tier1PSE.

Note that a descriptive report has not been provided for every Tier 1 PSE reported.

This database is a tool for learning and should not be considered a complete record of Tier 1 PSE in the upstream oil industry or the IOGP Membership.

AFRICA ONSHORE

DATE: 26 Dec 2023
COUNTRY: Congo
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the round, an operator noticed oily water spreading in the area of a tank's oil inlet lines. Loss of containment at the lowest point of a dead leg of about 70 metres, used in the past to feed another tank.

WHAT WENT WRONG?

Corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigation ongoing.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 23 Jun 2023
COUNTRY: Nigeria
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 1
ACTIVITY: Workover/Well services
MODE OF OPERATION: Well intervention / Well servicing
POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

The activity on well 2 started on 17 June with the equipment mobilization in location. On 18 June the job officially commenced on site. The Scope of Work was to carry out maintenance activities on Xmas Tree and ambient valve recalibration. Operation was performed with support of Slick Line service and downhole plug.

During the activity, the team observed the Slick Line and the upper sheave were not moving. In order to fix the issue, the Contractor Supervisor bled off the pressure in the hydraulic hose of the SL stuffing box through the dedicated hydraulic line, and proceeded to slack the cable at the bottom sheave to resume jar action This brought to failure the pressure equalization across the downhole plug and consequently led to the ejection of the same inside the lubricator which parted and jumped on air. The lubricator ejection determined the Slick Line cable tension and strong knockback. Consequently the lower sheave, connected to such cable, strongly impacted the IP's chest and chin. IP medical conditions immediately appeared critical. The event led also to a slight gas release.

WHAT WENT WRONG?

Rigless Programme developed PtW, Risk Assessment, Method of Statement, Isolation Certificate, TBT forms: all available in location, improvement needs have been observed in the procedures coordination and implementation.

- Lack of supervision.
- In presence of an issue (blocked Slick Line sheave), Contractor Supervisor decided to bleed off pressure in the hydraulic hose of the Slick Line stuffing box by pushing the small ball in the check valve seat allowing the hydraulic oil to bleed. This led to the failure of the pressure equalization and following lubricator's ejection. Operator did not stop the activity to re-assess the Risks.
- The Contractor's personnel competences were not previously assessed before starting the activity due to lack of Procedures. During the contractor's personnel interviews carried out in the Investigation process competence's gaps appeared evident for some crew members.
- Lack of knowledge on First Aid Operations and Emergency Communications. Night MEDEVAC Operations to be improved.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

While dealing with high pressure fluids ensure proper barriers always in place.

- In case the assessed workflow (as per method of statement) changes, proper risk assessment must be done and mitigations identified. It must be treated as management of change.
- Whenever you are planning a job, always take into consideration the operational context and the process conditions.
- All operations (routinely and critical ones) require competent person to be performed; their knowledge must be formally assessed prior to start any job.
- In all locations medevac drills must be performed and personnel to be adequately informed and trained about the existent procedures to be implemented.
- Proper communication devices in remote areas are fundamental to manage any emergency situation.
- Ensure safe rescue by medevac of the workers during day & night.
- Stop work authority to be always applied whenever required (anyone is entitled).

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Emergency Response Equipment and Systems

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 27 Nov 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

A loud noise was heard from HP separator area with hydrocarbon leak observed coming from the area. The Emergency Response Team mobilized to the area to douse the gas leak while the production team activated ESD and isolated the HP separators at the arrival manifold.

WHAT WENT WRONG?

Design / Management of Change:

The damaged 8" MP Line was under-designed.

It was connecting the HP separator to the MP header. Such line was installed to eventually divert the gas to header on MP mode in anticipation of depletion of the HP wells in future.

The line was not flanged immediately on the HP Separator Outlet but before the Inlet on the MP Header itself; thus, such configuration kept the line (about 20 metres) filled and pressurized at an operating pressure higher than its design pressure. Exact date of installation is unknown (evidence collected show its presence since 2002).

Alarm Management:

Only Rectifier 2 was in service but upon loss of electrical power, it went down. However, Rectifier 2 fault alarm could not be distinguished on the Power Management System (PMS) from alarm on Rectifier 1 (Rectifier 1 has been faulty overtime, with a constant active visual alarm on the PMS).

Training and Competence:

Due to loss of electrical power and UPS, there was a fail close of the three UV valves leading to HP gas header blocked outlet MP Compressors trip (with an alarm raised at the DCS) and consequently gas to flare. The UVs needed to be manually reset by the operator on site (to reopen). This action was not performed by any of the operators.

Preventive Maintenance:

There were 2 sets of PV valves connected to the HP flare, which were supposed to open for discharging all the amount of the excess gas. However, only 1 out of 2 sets of PV valves partially intervened to discharge the excess gas. The other set of PV valves did not respond.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Management of Change (MoC) process and related risk evaluation is crucial to ensure plant integrity.
- Life Cycle Information (LCI) is a cornerstone for safe and reliable plant operations. Process Safety / Asset Integrity documentation (e.g. P&IDs, Cause & Effect diagram, etc.) must be updated reflecting current plant configuration.
- Always ensure Safety Environmental Critical Elements (SECE) are well maintained, reliable and available, implementing and regularly updating the Asset Integrity Management Plan (AIMP).
- Always monitor process parameters both from DCS and with onfield checks.
- Always ensure that operators are properly trained on tasks to be performed and skilled according to their job position.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 06 Feb 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak observed along the Right of Way of a pipeline. Investigation revealed that the leak was from a hacksaw cut.

WHAT WENT WRONG?

Vandalism.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigation ongoing.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Acts of violence

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DATE: 07 Mar 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak observed on pipeline. Vandalization using hacksaw cut.

WHAT WENT WRONG?

Vandalism.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigation ongoing.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Acts of violence

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DATE: 22 Mar 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak observed on RUEL RoW. Vandalization suspected using hacksaw cut.

WHAT WENT WRONG?

Vandalism

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigation ongoing.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Acts of violence

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DATE: 06 Nov 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak observed on pipeline Right of Way due to vandalization.

WHAT WENT WRONG?

Vandalism.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigation ongoing.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Acts of violence

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DATE: 21 Jan 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas and Oil leak from well.

WHAT WENT WRONG?

Security Related (Third party interference).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: 08 May 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak observed on 28" pipeline ROW.

WHAT WENT WRONG?

Security Related (Third party interference).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

NONE: None: Unspecified

AFRICA OFFSHORE

DATE: 09 Jan 2023
COUNTRY: Angola
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Discharge to sea
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

An oil slick was observed on the sea surface, starboard side of an FPSO because of an upset in produced water treatment. Initially estimated quantity of LOPC was 130 litres of oil, based on observations on the day of the event. However, the quantity was revised on the day after, based on further observations and investigations on the incident root cause. The final estimation for discharged quantities was 10 cubic metres, based on aerial survey and matching with process estimates.

WHAT WENT WRONG?

Immediate causes:

1. An inhibition was implemented on oil/water interface level (LDAL) in a PW decantation tank without proper tracking, and consequently inhibition was left in place for 7 days. The oil/water interface level in the PW decantation tank therefore reached a critical level close to oily water booster pump suction. Oil was carried over to downstream hydrocyclone and surge drum, and a part of it was routed to overboard, leading to a spill.
2. Inhibition left in place has not been captured through the systematic daily review of PSS/ESD/F&G inhibitions (daily extraction from ICSS) as inhibited signal was part of PCS.
3. Two level transmitters (1 PCS, 1PSS) on PW decantation tank that should have triggered an alarm/shutdown turned out to be not working at the time of the event.
4. Two analysers that should have detected out of spec water were not working.

Main worsening factors:

1. Due to multiple changes of chemicals on the same day, actual reason for instabilities in PW treatment and off-spec water was not immediately identified.
2. Weather (East current) has contributed to a late identification of the origin of the sheen.
3. Several control loops are operated in manual due to Process conditions (e.g., amount of water to be processed significantly higher than during early phase of production) or due to poor reliability of transmitters (mainly LTs/LDTs). As such, flow of Produced water sent to PW decantation tanks is adjusted manually. This has contributed to the loss of level in PWT A.

Root causes:

1. Key information not tracked through shift CCR event log and at shift handover.
2. OIW Analysers selected technology not fit for purpose.
3. History of unreliable sensors combined with evolution of process conditions, leading to key regulations being under manual control.
4. Long delivery time to replace/repair analysers.
5. Long delivery time to replace/repair pumps and level transmitters.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. A list of alarms that are inhibited on DCS which details should be produced: – alarm tag/system/function – reason for inhibition – existing compensatory measure – whether inhibition of this alarm has been covered by weekly alarm management review.
2. Quality of CCR technician log should be improved: This log must include (not limited to): – all inhibitions implemented/removed during the shift including alarms/PCS/PSS/SSS/F&G/ESD/Package – all major changes of set points, – all change in chemical injections – all changes in well routing – all major events (instabilities). Key information must be clearly highlighted. This information must be discussed at shift handover.
3. OIW analysers to be repaired.
4. Reliability should be improved and if finally, reliability cannot be recovered, propose change of technology/SMIS/SMR for modifications.
5. Secure sustainably procurement and delivery of chemicals to avoid shortage and allow smooth transition between different type of products.
6. Remind all CCR technicians about the importance of screening periodically the views with a focus on inhibited signals (Signals Yellow squared).

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 09 Sep 2023
COUNTRY: Angola
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Discharge to sea
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During de-slopping operation, a valve spuriously opened, causing the slop tank pump discharge header to connect to the Crude Oil Tank transfer manifold.

Oil from transfer manifold contaminated the discharged slop water.

WHAT WENT WRONG?

- System card was found in short circuit.
- Valve activated and opened several times.
- Disfunction of the ODME (Oil Discharge Monitoring System), now registered in a

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Formalize, through a Procedure/Standing Instruction, the principles to allow Slop discharge to overboard while ODME is overridden.
2. Standing instruction to be issued to allow Slop discharge only if HV-28157 A/B are confirmed Locked closed.
3. Improve reliability of ODME: Review operating conditions, perform tests in different conditions (with and without slop water treatment package online), review adequacy of technology.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 14 Dec 2023
COUNTRY: Angola
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The cavity bleed valve located on the high-pressure compressor discharge shut down valve (SDV) to the gas injection system leaked resulting in gas release to the process area activating alarms and a process shutdown.

WHAT WENT WRONG?

Installation procedures do not include requirements for checks on valve fittings. Blow down valve was not maintained in accordance with company guidance issued in 2010. Emergency response training did not address best practices to manage and eliminate hazardous atmospheres.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Replace versa valve relays on blow down valves. Update installation procedures to check all valve fittings, including bleed valves. Develop response plan to manage responding to hydrocarbon releases and establish clear learning objectives for emergency response drill scenarios.

BARRIERS:

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 25 May 2023
COUNTRY: Congo
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During voluntary shutdown, operators heard an abnormal noise, and witnessed a cloud of gas between the separators. It was found that the gas recycling line from the compressor to the exchanger had burst open. Two operators were on the deck above the leak.

Due to shock wave, there was much damage to nearby lines and support. No one was injured.

WHAT WENT WRONG?

- Inadequate routing and quality review of risk assessment.
- No process expertise assigned to operational support.
- Low visibility of critical exposures.
- Criticality of discharge valve not recognised.
- Ineffective communication.
- Equipment overpressure detection design problem not considered, and hazard not analysed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Maintenance program and integrity review required. Pressurized Stop mode requires process review. Corrective actions on response to critical exposures need to improve (interdepartmental response to critical Asset Integrity exposures). Knowledge based decision making on complex systems need improvement, and recognition of critical exposures. Gas Compressor start and stop procedures need improvement to standardize method across shifts. Operating Instruction lacking in detail. Instruction on use and objectives of Operating Instruction and compressor start/stops needs improvement to ensure consistent understanding across shifts. Independent review of new leaks and defects needs improvement (Asset Integrity reporting against current risk picture).

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 21 Aug 2023
COUNTRY: Congo
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

During the preventive maintenance of an injection skid (hypochlorite), an accumulator was filled with Nitrogen. An overpressure led the accumulator to burst, and debris was projected around impacting two operators involved in the work.

The two employees (positioned at 2 metres and 6 metres) were injured: one on the arm and one on the face.

WHAT WENT WRONG?

Overpressure.

Use of a manifold (DBB) to inject Nitrogen without a pressure regulator connected to the cylinder. Non-compliance with the existing procedure. Poor risk assessment by stakeholders; Lack of supervision; Existing procedure not robust.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Initiate a site instruction to dissociate the accumulator recharging permit (conditional operation) from the pump preventive maintenance permit (planned operation).
2. Draw up a specific procedure for recharging anti-pulsation balloons, including the use of means adapted to refilling pressures (10 barg nitrogen network and loading pressure 5.9 barg; instead of a 200 barg nitrogen cylinder).
3. Raise staff awareness of the need to comply with procedures attached to work permits.
4. Hydraulics team should be trained again in the basics of pressurized systems.
5. Ensure that the site is equipped with a certified pressure reducer (controlled integrated valve) according to good practice.

6. Define a pressure reducer storage area known to all people involved in the activities.
7. Raise staff awareness of the importance of insulation on the double sluice gate.
8. Include the phrase "critical tasks may not be performed at the end of the day" in the CP identifying critical tasks on the site.
9. Raise staff awareness of the importance of good communication and supervision during task execution.
10. Draft and disseminate a REX.

BAR!RIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 18 May 2023
COUNTRY: Nigeria
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At about 12:30 hrs, while attempting to circulate a dual producer well with fresh seawater via coil tubing as part of activities to perform well kill operation in response to an initial leak, there was a loss of containment of hydrocarbon gas through a 20-inch conductor casing of the well.

WHAT WENT WRONG?

Operations Procedures inadequate, Controls or preventive systems inadequate, Maintenance Procedures inadequate.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

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DATE: 30 May 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At about 19:30 hrs, Marine Breakaway Coupling (MBC) on a cargo hose string was activated during the loading of a crude oil export tanker. The cargo hose string was separated, with one end connected to the manifold of the export tanker and the other end connected to the single point mooring, and by design, releasing some barrels to water on activation of the MBC.

WHAT WENT WRONG?

n/a

CORRECTIVE ACTIONS & RECOMMENDATIONS:

n/a

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 24 Feb 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Personnel perceived smell of hydrocarbon gas during Daily Site Sweep around a Water Ballast Tank (WBT) undergoing repairs. A leak point was traced to a spot identified on the bulkhead between the WBT & Crude Oil Tank (COT) with a 12mm diameter leak spot size.

WHAT WENT WRONG?

Immediate causes:

Wall through defect due to severe corrosion on bulkhead between crude oil tank and water ballast tank.

Root causes:

1. Inadequate project works execution with failure of Quality Assurance/Quality Control (QA/QC) process.
2. Failure to follow Procedure: Management of Change of repair strategy from use of insert plates to painting and blasting of corroded sections was done but not approved by Technical Authority.
3. Inadequate inspection of bulkhead as further Non-Destructive Test (NDT) required for corroded sections was not done after previous repair.
4. Weak Project Organization with insufficient clarity on roles, responsibilities and chain of command leading to improper level of technical validation of inspection reports.
5. Manual override (inhibition) of gas detection safety system inside the WBT was done without following procedure leading to late detection of hydrocarbon in the WBT.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Reinforce inspection team onsite and support organization to ensure right level of technical validation of inspection reports.
2. Reinforce change management procedure where there is a change of scope.
3. Re-enforce Project organization/chain of command on site for technical approvals.
4. Implement workflow for inspection/repair/work follow up with clarity in contractor/company QA/QC reporting and validation steps.
5. Implement milestone review, QA/QC and inspection, test plan and validation of repairs by all parties.
6. Re-enforce implementation of full commissioning process culminating with "Line-Walks" and Ready-For-Start-Up (RFSU) by RSES for all completed works before coming into service.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

DATE: 05 Jul 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Operator observed increase in volumes discharged to flare and determined that PSV pilot sensing line separated from PSV and 3/4 inch bleed valve nipple separated from weldolet downstream of the PSV.

WHAT WENT WRONG?

Overtightening of sensing line tubing connector led to crack which propagated over time. Small bore connection downstream of PSV was not braced in accordance procedure updated after initial design and start up.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Update applicable procedures to capture correct method for installing threaded fittings and train personnel on update. Prioritize and implement bracing of small bore piping in vibrating service.

BARRIERS:

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 15 Nov 2023
COUNTRY: Nigeria
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Approximately 17 hours into a crude oil offload operation from an FPSO to a cargo tanker, during the crude oil offloading operation a low-pressure alarm activated on offloading pumps. Oil was detected at the sea surface.

WHAT WENT WRONG?

Export hose suffered a full-bore failure; Unexpected disconnection of flanged nipple from 'first off the buoy' section.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigation ongoing.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

ASIA / AUSTRALASIA ONSHORE

DATE: 26 May 2023
COUNTRY: Australia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Turnaround
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (not at intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We recognize change.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

An internal combustion event occurred inside the air assist system of the Stage 1 warm wet flare which resulted in two loud bangs.

WHAT WENT WRONG?

1. Design Review: The risk of hydrocarbon ingress into the air assist system was not adequately considered during the HAZOP post the change in the flare system operating strategy.
2. Corrective Action – Needs Improvement: Risk of hydrocarbon ingress into a flare air assist annulus was identified in a prior event with lessons potentially not clearly articulated.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Standards: Before the commencement of a task, familiarize yourself with primary and secondary muster points.
2. Risk identification: Vendor engagement during front-end design and safety reviews/HAZOP specifically for novel or complex systems is important.
3. Communication: Operating procedures to provide key vendor parameters and a clear line of sight to operational risks.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 02 Apr 2023
COUNTRY: Australia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Heat exchanger
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At approximately 01:30CST, a Heating Medium (HM) leak was reported on LNG Train 1. The leak was detected by an Area Operator during the completion of routines and rounds on the train. LNG Train 1 was safely shutdown, ERT were mobilized to control and contain the leak. The cause of the Loss of Containment (LOC) was identified as the full rupture and separation of a tube on the HM air cooler bundle.

A previous LOC occurred on 15 January 2023, where a tube from the same air cooler was cracked (not fully ruptured). Metallurgical examination identified the cause of the crack as being progressive, corrosion assisted fatigue cracking mechanism. The fully ruptured tube from the 2 April event, the visual signs indicate the same failure mechanism.

WHAT WENT WRONG?

- HM surge drum design causing excessive nitrogen entrainment from the HM surge drum into the HM circuit, leading to N2 accumulation at the highpoints and gas locks in some bundles of the HM cooler. The bundles with no gas lock then experience high HM flow, with high velocities and vibration on the top row tubes.
- No operational venting points were provided by design at the HM cooler, which lead to the accumulation of entrained nitrogen at the high points of the HM cooler.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Increase HM Surge Drum levels to reduce N2 entrainment.
- Regular venting of trapped N2 from the installed venting points and install an automated venting system.
- Implement new internals in the HM surge drum (e.g., inlet device, inlet/outlet piping configuration) to minimize N2 entrainment in the HM circuit.
- Perform regular monitoring with thermal and vibration cameras and digital camera inspection on the heating medium circuit.
- Review the heating medium circuit process controls to minimise flow and thermal cycling.
- Study the installation of a downstream isolation valve on HM cooler to significantly reduce the boundary isolation requirement on the HM cooler.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 12 Mar 2023
COUNTRY: Malaysia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

LTI – Hot process water splashing during vent de-blinding.

WHAT WENT WRONG?

Burn injury due to inadvertent release of steam

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Line of fire activities that are proven unnecessary to address operational challenges. Continuous review and improvement of existing routine operational procedures to capture lessons learned.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment

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DATE: 03 Oct 2023
COUNTRY: Malaysia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

Non-Emergency Medevac: Chemical Splash on Personnel Face & Shoulder.

WHAT WENT WRONG?

Equipment failure. Dislodged 12 mm small bore stainless steel tubing.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Line of fire activities that are proven unnecessary to address operational challenges. Continuous review and improvement of existing routine operational procedures to capture learnings.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

ASIA / AUSTRALASIA OFFSHORE

DATE: 27 Oct 2023
COUNTRY: Australia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Turnaround
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Tier 1 – HCl Flange Leak (240L) Floor Deck Port-side STG Auxiliary Room.

WHAT WENT WRONG?

Release of material from flange

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Leak caused by poor quality, non-spec PTFE lining on the SS316L spectacle blind.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 04 Apr 2023
COUNTRY: Malaysia
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Drilling
MODE OF OPERATION: Drilling
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas kick at exploration well.

WHAT WENT WRONG?

Well collision during drilling.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Sustain emergency drills and conduct as required to ensure all personnel are familiar with their respective roles during emergencies.
2. Ensure the fire and gas detection system is healthy and functioning as intended.
3. Ensure Shutdown System (SDS) is as per Approved Cause and Effect Matrix (CEM) and functioning as intended
4. Continue to be vigilant and stay focused when drilling is in operation
5. Maintain close monitoring of active drilling activities by maintaining close communication with the rig floor.
6. Ensure emergency protocol in terms of notification and response is well established when involving multiple entities at a location.
7. Ensure simultaneous activities are well-managed and coordinated through effective Specific Instructions for Simultaneous Operations (SISO) documentation.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 14 Sep 2023
COUNTRY: Malaysia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Subsea: Subsea pipeline/flowline
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Hydrocarbon was observed released through a small opening on flexible riser close to the flange connection to Pipeline End Manifold (PLEM) elbow.

WHAT WENT WRONG?

Flexible riser fatigue.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Be more vigilant and alert on the surrounding changes arising from operational activities and to immediately report any abnormalities.

2. Review the adequacy and frequency of inspections, testing and preventive maintenance in ensuring equipment integrity.
3. Ensure emergency protocol in terms of notification and response is well established when involving multiple entities at a location.

<<No Barriers Allocated>>

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 08 Jul 2023
COUNTRY: Malaysia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

While performing FCV stroke check, suddenly a pin hole leak happened from nearby system, causing sulphuric acid mixture splashed on a cable tray and deflected to the person's face.

WHAT WENT WRONG?

Pinhole leak at chemical tubing.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Be alert on the surrounding system especially in the chemical area.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE ONSHORE

DATE: 02 Feb 2023
COUNTRY: Hungary
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A deep pump oil well, started up from the left. Due to the increased yield, a tank which was under filling filled up shortly after the maximum level signal was received, and 5 cubic metres of stratified water and 5 cubic metres of crude oil spilled through the tank's breather.

WHAT WENT WRONG?

Communication.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

More frequent inspection, installation of high-high alarm.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 11 Mar 2023
COUNTRY: Hungary
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The oil pipeline 350 metres from the main collector's fence was punctured underground.

WHAT WENT WRONG?

Unintentional breach of rules/regulations.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Update P&IDs, check the out of use pipeline sections, review of technological descriptions.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 11 Mar 2023
COUNTRY: Hungary
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The oil transport pipeline was punctured underground 30 metres from the fence of the Main Collector.

WHAT WENT WRONG?

Unintentional breach of rules/regulations.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Update P&ID, review of out of use pipeline sections, review technological instruction.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 08 May 2023
COUNTRY: Italy
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Tanks and Sumps/Pits: Sump/pit overflow
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

At approximately 19:00, there was a vapor leak from the triethylene glycol regenerator. The aqueous portion, due to the operating temperature (~180°), instantly vapourized, creating a visible white cloud even from the outside. The leak occurred following a sudden rupture in the accumulator/reboiler intercommunication line. Central operators shut down the unit and closed the valves, stopping the leak (Event classified as PSE Tier 3). The rupture occurred on the branch, downstream of the valve, normally closed, used for draining the reboiler to the accumulator. Upon further analysis, the valve body was also found to be punctured. Condensate water accumulated inside the containment basin. There were no injuries to personnel or damage to the environment.

Subsequently, in order to carry out line restoration operations and valve replacement, the central personnel decided to immediately drain the equipment, also considering the out-of-service status of the other regenerator (under maintenance since March 2023) and the possibly reduced quantity of already regenerated TEG stored in the tanks. The operations began by draining the accumulator (lower body of the regenerator) via the dedicated 2" line, sending the contents (still hot, as the equipment had been shut down for only a couple of hours) into the sumps. At approximately 21:00, during the drainage activities, a release of hot vapour mixed with drained glycol occurred from the sumps, affecting the chief central assistant present at the workplace as the Observer.

WHAT WENT WRONG?

- Structural failure of the interconnection branch, with emission of steam and glycol [associated with Tier 3 event].
- Activity carried out without allowing the drained fluid to cool down [associated with Tier 1 event].
- Failure to assess the risk of semi-liquid discharge presence [associated with Tier 1 event].
- Backup regenerator was under maintenance [associated with Tier 1 event].

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Evaluate the necessary time and perform the necessary checks before starting the drainage operation.
- Consider the possible presence of other liquids in the semi-liquid water discharge systems and the potential risks.
- Ensure the presence of an operational instruction for carrying out the activity.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 28 Dec 2023
COUNTRY: Italy
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Drain
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

A production operator, during a routine check, noticed a slight drip of hot regenerated amine from the cap placed on the manual drainage valve of the filter at the suction of the pump. After requesting the intervention of the mechanical service team, the cap in question detached from the filter, causing the sudden loss of the amine solution at about 110°C. The spilled liquid came into contact with the operator's forearms, equipped with the PPE supplied. The operator was promptly accompanied to the infirmary and after receiving the appropriate medications, returned to normal work activities. The operator continued his work shift (night shift on 29 and 30 December) without encountering any particular criticalities. On December 31, 2023, at around 05:30, the operator felt severe pain in his forearms and went to his home at the end of his shift. In the afternoon, due to the persistence of the pain, the operator decided to go to the emergency room for tests where he was diagnosed with a second-degree burn to the forearms with an initial prognosis of 7 days.

WHAT WENT WRONG?

The operator, while promptly reporting the leak of liquid from the filter to the control room control room, remained within the area of potential impact without, however, being able foresee the negative development of the release.

No change management (MoC) process for the modification of the filter drain type.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

LINE OF FIRE: All personnel must maintain a high level of attention during work activities, especially when these involve substances chemicals and/or high temperatures. One must constantly pay attention to their own work and, if necessary, stop/review it if, as the activity progresses, potential hazards related to potential dangers related to the "Fire Line".

MOC: Plant/organizational changes must be identified, analysed and authorised. On the other hand, Contractors who carry out installations and/or restoration of items and/or components must always comply with the project and installation specifications. If inconsistencies emerge, it is essential that the company reports the anomalies to the Company.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 05 Jun 2023
COUNTRY: Romania
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Between 4 and 5 June 2023, an oil-water spill occurred in an agricultural area 500 square metres downhill from buried pumping pipeline connecting park to tank farm. The Loss of Primary Containment (LOPC) event lasted for around 20 hours, until Monday, June 05th (22:00 Hr) when the pumping was stopped. Production measurement at the tank farm and calculations indicated an estimate of 200 cubic metres gross oil-water volume that were lost (85 tons oil and 100 cubic metres produced water). The pipeline operating pressure was approx. 7.5 barg.

WHAT WENT WRONG?

The Root Causes:

- Pipeline loss of metal due to internal pitting corrosion lead by under deposit corrosion most probably triggered by high presence of Sulfide Reducing Bacteria (SRB).
- No corrosion prevention / mitigation (Corrosion never inhibited since the pipeline startup more than 20 years ago, no internal cleaning (pigging), no biocide injection).

- Non-Destructive Testing through wall thickness measurement is limited to some spots that do not give information on the whole pipeline integrity condition.
- The soil permeability in the relevant area is such as not to allow the surface appearance of the spill in the proximity of the pipeline.
- Detection production loss is based on working instruction that in the best case can be identified after more than 8 hours in case the leak cannot be easily observed by the patrolling.
- No online process information on the critical parameters as pumping flow at inlet of the Tank farm.
- Delayed decision-making process to stop pumping.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Work instruction to be modified in order to measure stop pumping when it is confirmed that the production loss is not due to specific well(s) production that have declined or stopped.
- Install gross production flow meter at the tank farm manifold of each inlet line or at least on the inlet tank's pipe.
- On the lack of possibilities to implement corrosion prevention mitigation, replace the sections with temporary clamps on the carbon steel section or replace the whole carbon steel section with non-metallic material (1.6 km length).
- Install overpressure and low-pressure protection at the pumping system of the upstream facilities.
- Install check valve at each pipeline getting in the tank farm.
- Re-assess pipeline risk in pipeline integrity management software to understand if the current risk after the incident is high, and proceed with measures to include in replacing program.
- Implement pipeline ICE pigging to remove deposits and reduce the likelihood of Microbiologically Induced Corrosion (MIC).

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

DATE: 05 Jul 2023
COUNTRY: Romania
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

While the contractor team was replacing a valve from park manifold, salt water and gas were released between valve flange and manifold flange. The worker tried escaping the working area, lost balance and injured his leg. As the LWDI occurred while reacting to a process safety event, this qualifies as Tier 1 incident.

The incident happened while the maintenance crew performed the dismantling of the old pipeline of the Well, which has stopped production for 6 years, at the end part of the installation which leads to the separators of the gathering parc. This work was necessary for the preparation of the new pipeline construction that was planned to be executed by another contractor. The old pipeline dismantling activity started two days before the incident. The access routes were not finalized, the metallic footbridges were not connected and there were no handrails, therefore a continuous and safe route to the facility/equipment or to the muster point did not exist.

WHAT WENT WRONG?

- Poor application of internal regulations: Permit to Work, Plant and equipment isolation.
- Prioritization of maintenance activity depending on production opportunities.
- Poor risk awareness and management for the aspect of pressure and hydrocarbon presence in equipment.
- Company and Contractor have high risk tolerance and, their activity is influenced by routine.
- Inefficiency of training for emergency situations.
- Workplace Risk Registers not adapted to the actual configuration of the facility, to reflect specific hazards.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Reinforce knowledge of Permit to Work (PTW) procedure application for all Company and Contractors' employees.
- Apply consequence management for the cases where work starts without the validation of PtW on site.
- Improve the process of preventive maintenance.
- Implement properly the equipment isolation process, including the ''extended period isolation" requirements.
- Implement training plans on P&ID utilization.
- Updating workplace risk registers by assessing and treating the current risks at the operational site.
- Review the Emergency Evacuation Plan and improve its implementation (e.g. exercise its application).

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/ materials/products

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/ materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

EUROPE OFFSHORE

DATE: 29 Nov 2023
COUNTRY: Norway
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Relief, Vent and Discharge Systems: Discharge to sea
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

After start-up of an offshore production facility following an unplanned shut-down, oil was observed at sea. The oil discharge was stopped by manually closing the degasser outlet valve. It turned out that reverse oil flow from the second stage separator to the hydrocyclones and further to the degasser and out to the sea had taken place.

WHAT WENT WRONG?

The main cause for this incident to happen was that a check valve preventing reverse flow had not been installed. Furthermore, the online oil-in-water measurement device was not operational at the time preventing detection of oil in the hydrocyclones.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

A design review will be carried out to find the best solution to prevent reverse oil flow from the separator to the hydrocyclones. Before implementing any changes, a HAZOP involving a multidisciplinary team with the relevant skills will be carried out. Procedures for oil-in-water measurements to be revised.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 22 Apr 2023
COUNTRY: Norway
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

Prior to the installation of the new hook-up spool, a blind lid on a new flowline to well was to be split and removed. Confined pressure was not detected, and pressure was therefore not drained before the split was performed, resulting in lifting the blind lid (34 kg) 1.6m up in the air – this hit the operator in the face on the return. 2.4 kg HC gas escaped, resulting in gas detection, 2 sensors activated – this lead to activation for the deluge system and ESD. One person in the work team was seriously injured.

WHAT WENT WRONG?

Immediate causes:

- Splitting was carried out with pressure in the flowline.
- The work permit was activated without the correct plan for isolation (ICC) for the work being prepared, linked to the work permit and put in place (live).
- Those involved incorrectly perceived that it had been verified that there was no pressure in the flowline.

Most important underlying causes:

- Drifting everyday practice in processing work permits, irrelevant ICC linked to work permit, and information in operational overview that was not taken into consideration.
- Verification of depressurized system at one verification point, and with incorrect verification method.
- The mechanics loosened the clamps in accordance with requirements, without getting any impact on the gas detectors, and therefore perceived that the system was depressurized.
- High operational tempo, quality in Work Order plans, and resource coordination between several interfaces resulted in strain on capacity for the operations discipline.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure and verify integrity of the management of isolation certificates (ICC), through training and verification of methods for testing of depressurization and HC-detection, and correct ICC list linked to Work Permit – secure operational capacity for planned operations – mentoring/on-the-job training to increase operational competence – strengthen compliance of work requirements through self-assessment and peer review/verification.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 15 May 2023
COUNTRY: Norway
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas-alarm on tank deck. When the 3 gas detectors no longer indicated gas, the search & rescue team entered and found the leak to be from the entry hatch of crude oil storage tank 2 on the port side. The incident was caused by weakening of the hatch gasket.

WHAT WENT WRONG?

Defect or technical failures on component/system/plant – wrong type of material(s) used.

Comments:

- Wrong sealing type.
- The risk of weakening of hatch gaskets was known, and a plan for repair was linked to the maintenance programme over several years.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Conduct a lessons learned meeting after gasket replacement with involved personnel.
- Update the maintenance programme with leak testing after opening of tanks.
- Check gaskets on all hatches against storage and slop tanks on the tank deck.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Execution of activities

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

MIDDLE EAST ONSHORE

DATE: 05 Dec 2023
COUNTRY: Iraq
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Low level on 1st stage separator triggered the closure of oil outlet valve.

Oil level further increased in the separator up to High Level that led to shut down by closure of crash valve upstream separator.

WHAT WENT WRONG?

Well shut down valve is inefficient, remote activation not available.

Design pressure of the flowline > 400b. Pressure before rupture 130b.

Remaining thickness of the pipe at the failure point (welding) 5 mm vs 15 mm initial design.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigation ongoing.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 14 Apr 2023
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Heat exchanger
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A condensate leak was identified from De-ethanizer Reboiler Bundle E-225-C1. In the initial stage, condensate dripping was observed and a drip tray placed under the leak and a barrier installed around the area.

The operations team observed and monitored the leak. When the leak rate increased, the Operations Supervisor informed HSE, Emergency Response Fire Team, and HOD for further actions.

Further investigation revealed that during efforts to manage hydrates in the Cryo systems, De-ethanizer Column (T-521-C1) was subjected to pressure above the Maximum Allowable Working Pressure (MAWP) of 550psig, and outwith the maximum range of the pressure reading instrumentation (567psig), remaining in this state for a period of approximately 9.5 hours.

WHAT WENT WRONG?

1. Lack of adequate overpressure protection on T-521 (De-ethanizer Column).
2. PAH-521A disabled in SCADA.
3. No formal procedure/process in place currently for the management of hydrates within the cryo system.
4. High dew point when Mol Sieve bed V414 exceeded adsorption time by 4 hours.
5. Prolonged operation at high flow rates and high dew point.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Engineered measures to prevent overpressure of the expander, de-ethanizer system to be implemented at the earliest opportunity.
2. Pressure transmitter (PT-521A) to be used to provide an input for an executive action that isolates vulnerable sections from upstream pressure sources.
3. Reinstate PAH-521A at the earliest opportunity and develop an approved documented alarm response.
4. Current methods used to 'warm' the system after the introduction of hydrates to be ceased with immediate effect.
5. Review critical process parameters and how they might be better monitored by control room personnel to ensure the plant is maintained within the safe operating envelope.
6. Review reinstatement of dew point trip (ME-416-C1/C2) to establish if this can be achieved with sufficient reliability of the instrumentation.
7. Develop formal procedure for mol sieve operation including normal, abnormal, and emergency scenarios.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 11 May 2023
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 15:35, C-131 tripped. When an operator started the unit back up, the system then opened the suction valve. At this point the suction PSV lifted and gas was released to atmosphere and directed toward top of V-414-C2 which damaged insulation, which fell to the ground. When it happened, an operator was near to that area and witnessed the damage.

The failed PSV 131 was submitted to the PSV mobile bench test. The "as received pre-pop bench test failed. The PSV passed below the PSV set point. The PSV was dismantled, and it was confirmed that the PSV elastomeric disk seal was damaged, thus could not hold the set pressure.

It was also noted, a minor deposit, fouling dust accumulated between the seal and the disc which could lead to a PSV not meeting the previous set pressure.

WHAT WENT WRONG?

Enforcement of SPP not effective: well-crafted SPP were in place, but their use was not properly enforced to the extent necessary; the operator was still in training period but assigned as Compressors operators.

1. Skill level/competency: Assessment of required skills or competency not effective: The employee was still in training period,
2. Skill level/competency: Lack of coaching on skill: insufficient coaching provided for the operator.
3. Training/knowledge transfer: Training material not recalled: The process operator attended class & on job training, but he did not recall it.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Only Operators with signed off Compressor competencies and experience to be area owners of the Compressors. No untrained personnel are to work in the compressor area unsupervised.
2. Area owner to coach new operators in the SSP for the compressor area. Training and competency to be supervised and signed off once practical assessment of knowledge and skills is undertaken and deemed suitable.
3. PSV line elbow to be cut and vertical to atmosphere

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 04 Jun 2023
COUNTRY: Kuwait
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from a 6'' underground line coming from oil wells and going to Gathering Center. The line was isolated. The total volume spilled was 75 bbls.

WHAT WENT WRONG?

Underground flow line leak – Corrosion/ erosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 15 Feb 2023
COUNTRY: Kuwait
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from the flange of 34" aboveground line coming from central mixing manifold and going to Tank Farm. The line was isolated. The total volume spilled was 20 bbls.

WHAT WENT WRONG?

Above ground flow line leak – Corrosion/erosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 06 Feb 2023
COUNTRY: Kuwait
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude oil leak from 6" underground line going to Early Production Facility. The line was isolated. The total volume spilled was 10 bbls.

WHAT WENT WRONG?

Underground flow line leak – Corrosion/ erosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Periodic maintenance/inspection/testing.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 03 Dec 2023
COUNTRY: Kuwait
FUNCTION: Production
NUMBER OF DEATHS: 2
ACTIVITY: Unspecified – other
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

INCIDENT DESCRIPTION:

On the day of the incident, the gathering center was under total shutdown. The job was to replace the corroded 42" tank vapor gas piping. The activity was performed by cold cutting by the contractor. While cold cutting, an explosion and fire occurred that resulted in two contractor fatalities, 12 LWCs, and 1 minor injury. All emergency responders attended. The fire was extinguished by fire crew. The employees were transferred to medical facilities by ambulances. The deceased person was taken by forensic medical department.

WHAT WENT WRONG?

Workers in the line of fire.

Underestimated the hazard of work location and activity.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Ensure Method Statement for Critical Activities includes all potential hazards and precautions to eliminate/mitigate hazards and ensure compliance.

Ensure close follow-up between all the maintenance staff and contractors during facility shutdown.

<<No Barriers Allocated>>

<<No Causal Factors Allocated>>

DATE: 28 Jun 2023
COUNTRY: Kuwait
FUNCTION: Production
NUMBER OF DEATHS: 1
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

When a Production crew was conducting routine well monitoring and troubleshooting, they noticed a spill from a nearby pipeline. The crew approached the spill site to investigate. Upon arriving at the site, the crew saw the water disposal pipeline had failed, and a pinhole was sending high-pressure produced water into the desert sand, creating a five-meter wide and two-meter-deep whirlpool. During this observation, one of the workers slipped into the newly formed whirlpool. Crew members made several attempts to rescue him and were finally able to retrieve him from the water. One of the rescuers went into the hole and had to be rescued without any further harm. Emergency Medical arrived on site to begin advanced life-saving techniques but were unsuccessful.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 07 Nov 2023
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Temporary
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

An officially declared community evacuation or community shelter-in-place

INCIDENT DESCRIPTION:

A complaint was received of high SO2 emissions affecting a neighbouring project. High SO2 emissions were expected as part of the temporary

operations, defined in "Onshore Plant Operational Guidelines During Sea Water Headers Failure," put in place to attempt to address an underground seawater (SW) GRP leak.

WHAT WENT WRONG?

TGTU bypass resulting in high SO2 emissions.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Potential risks involving neighbouring facilities should be shared, documented, and communicated with all involved entities.
2. Enhance communication through the approved protocol of the city. The communication shall ensure two-way communication and include health and safety impacts, if any.
3. MOC risk assessments to consider the potential impacts on neighbouring facilities (note: including facilities under construction).
4. Conduct regular drills tackling the incident scenario (involving other affected operators).

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 06 Oct 2023
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Equipment: Pig launcher/receiver
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The station tripped on the activation of the gas detectors due to the gas leak from the rupture of the 4" drain line of pig launcher. This happened during the pressurization of the 18" Gas Injection Header at 100 Bar after the pigging job was completed. The plant was depressurized, and the pig launcher was isolated. The 4" ruptured drain line spool was removed and blinded. The plant was made safe and started back.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

NONE: None: Unspecified

DATE: 23 Jul 2023
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Propane Leak from C3 compressor discharge transmitter tubing.

WHAT WENT WRONG?

High vibration resulted in the compressor discharge transmitter tubing detaching.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Modify layout of impulse tubing and location to minimize vibration.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 27 Oct 2023
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Equipment: Pressure vessel
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Flammable gas leak from top flange of expander feed separator pressure vessel.

WHAT WENT WRONG?

Manufacturing/installation issues.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 11 May 2023
COUNTRY: UAE
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

During tie-in activity involving a 20-inch production header and a 4-inch test header, a fire ignited that resulted in injuries to two personnel. Immediately emergency shutdown was activated, and the fire was extinguished by the tactical response team. The injured persons were evacuated to site clinic and were later referred to hospital for further treatment.

WHAT WENT WRONG?

- Inadequate implementation of PSP.
- Hot Work Permit was chosen instead of Critical Work Permit and subsequently associated Formal Risk Assessment for the activity was not conducted as JSA level 2 was utilised for the PTW.
- Non-proved isolation was utilised (single valve isolation – without bleeding).
- Inadequate planning or risk assessment performed.
- Unplanned draining of 4-inch test header without appropriate WMS/FRA & Energy Isolations.
- Inadequate enforcement of PSP.
- Non ATEX Certified Pneumatic driven hydraulic torque wrench and sparking spanners and tools (except hammer) were used on 4-inch flange in the hazardous area.
- Draining, flushing, and purging were not undertaken.
- Inadequate assessment of needs and risks – the available hydrotest vent was not utilised to conduct the appropriate flushing of 4-inch test header.
- Inadequate technical design – inadequate draining due to non-availability of low point drain in the header.
- Inadequate identification of worksite/job hazards – the crew moved to SIMOPS with 4-inch test headed spade rotation work without fully securing the 20-inch production header flange (i.e., full tightening including all bolts).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Commitment and accountability

Management System Element Barrier Failure: Policies, standards and objectives

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 03 Jun 2023
COUNTRY: UAE
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Upset
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

At 13:40, as part of troubleshooting of operation upset in utilities area, excess low-pressure steam was being discharged through a sky vent. During the process condensate carried over with the steam to the sky vent. While an operator was observing the release of steam from the 3rd level platform, steam discharged and steam condensate splashed from the sky vent, causing burns on his back.

WHAT WENT WRONG?

1. Inadequate Technical Design.
2. Inadequate Development of PSP.
3. Inadequate Management of Change.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Review the adequacy of the associated pipeline, drain and steam trap to prevent condensate accumulation.
2. Review the standing instruction to address all hazards, risks, and controls.
3. Review the change in operation philosophy to open Pressure Valve based on LP steam flow above 420 T/Hr and document the change if required by a technical Management of change process

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Overexertion or improper position/posture for task

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST OFFSHORE

DATE: 01 Sep 2023
COUNTRY: Qatar
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Drilling
MODE OF OPERATION: Unspecified drilling
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Control Room reported to the Well Head Maintenance representative at the rig that gas alarm (Toxic and LEL) received from gas detectors and followed by WHJ trip. Surveyed the WHJ under 30 minutes using Breathing Apparatus set with buddy system, and noticed gas leaking from A-Annulus pressure transmitter ½" SS-tube dislodged from DBB male connector. Isolated DBB and secured with ½ OD plug. Also observed during the investigation that a well X-mas tree was significantly moving during the bad weather (swell 5'-6' at the time of incident).

WHAT WENT WRONG?

1. No Common Standard on movement control design between projects, Drilling and Operations.
2. Lack of hazard recognition/use of take time.
3. Poor design standard for small tubing connection/layout.
4. Lack of interface management with implementation of Learning From Incidents.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Develop a common standard for the control of movement of conductors and platforms in coordination with Operations.
2. Install STOPPERS onto all wells requiring Stoppers prior to being put into production.
3. Small instrument tubing connection and layout design to be specified in design requirements.
4. Ensure LFI actions are shared with all parties to cover existing and future requirements.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

DATE: 12 Sep 2023
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

WHO recorded 1st casing 4 bar and depressurized to 1 bar for W/L. W/L was scheduled to install RV plug in 1st casing opposite of G/L side. Blind flange was removed by WHO to install RV plug. RV plug found installed already when flange was removed. RV plug was removed by W/L without clearance while depressurization was ongoing, resulting in uncontrolled control gas/oil release. Initiated ESD locally from WHJ and secured wells.

WHAT WENT WRONG?

RV plug was removed by W/L without clearance while depressurization was ongoing, resulting in uncontrolled control gas/oil release.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Need for improved communication between team members.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 10 Nov 2023
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

While taking Wellhead data, a gas leak was identified from downstream of a Gas Lift line, Offshore. The pipe section was immediately isolated and depressurized. Inspection revealed the leak was due to a pinhole of approximately 5mm diameter (external corrosion) which led to estimated leak rate of ~ 1000 kg/hr at operating conditions.

WHAT WENT WRONG?

External corrosion led to pinhole leak.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 08 Jun 2023
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Subsea: Subsea pipeline/flowline
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A helicopter pilot noticed a large bubbling of production gas at the sea surface, offshore, north-east of a Production Station. A boat confirmed the gas bubbling with no traces of oil. Using an ROV, the Diving Support Vessel located the source of the leak at a subsea pipeline, related to a flowline transferring production to the production station. The well was then closed, the line depressurized, and a Piping clamp installed.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 25 Jul 2023
COUNTRY: Qatar
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

General alarm was activated based on confirmed toxic gas detection at a Production station. Shortly after, personnel were instructed to don 15 minutes BA sets and proceeded to the designated primary muster stations as per relevant emergency procedure. Subsequently, toxic gas high alarms (2ooN) were reported in three adjacent platforms. A short while later, backup fire and investigation team confirmed oil station flare was in flameout condition. Toxic gas alarm was activated in the Platform and initiated automatic closure of the air intake dampers. Found Oil Station Flare went off due to suspected pressure fluctuation on LP Flare header and HP flare header pressure controls, thereby creating disturbance of flare gas flow rate. No PSV relieve or Blowdown activation was reported/ recorded prior to the flame off. An hour later, all clear mustering alarm announced and personnel dismissed from primary muster stations and returned to their respective routines. CCTV camera playback later showed occurrence of the flame-out at around 00:30. Hence, flare flame off duration was approximately 4 hours.

WHAT WENT WRONG?

1. Unavailability/non-functional oil station flare pilot re-ignition and pilot monitoring systems.
2. Inadvertent closure of pressure control valves to HP flare and to MP flare for a duration of approximately 4 minutes shortly prior to the flame-off occurrence.
3. Maintenance Management: Unloaded Compressor (1st and 2nd stages) due to outage.
4. Asset Integrity Management: Inadequate rectification of known degraded and deformed HP/MP/LP flare tips.
5. Organization and Human Factors: Inadequate manning of Control Room during night shifts.
6. No risk assessment was performed for the non-functional Pilot Ignition System and mitigating actions were not identified.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Initiate Risk Assessment and issue a Continuous Case for Operations due to unavailability of flare ignition system.
2. Install independent flame-off detection system as a back-up to thermocouple-based Pilot Flame Detection System and CCTV system for early detection and warning of flare flame off events.
3. Perform tuning of the HP/MP header PCVs for quick response to avoid sudden closure/opening.
4. Ensure at least Control Room Operators at all times to avoid overburdening of tasks in the CCR.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 27 Jul 2023
COUNTRY: UAE
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Relief, Vent and Discharge Systems: Drain
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

During the start-up of LNG Train 3, condensate level in flash drum (operating at 8 barg) and while operators were opening the drain valve, suddenly condensate steam released from the underground trench. Both operators were exposed to steam and sustained burns of varying degrees.

WHAT WENT WRONG?

Defective Equipment (Drain line suffered from severe corrosion and metal loss due to corrosion (Metal thickness Design Vs Actual 5.54 mm Vs 0.86 mm (localized)).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Prioritize inspection and maintenance of all drain lines within the plant.
2. Identify and rectify any deteriorated or compromised drain lines to ensure they are in a safe and operational condition.
3. Revise the inspection procedures to include drain lines and other seemingly low-risk components, especially when aging or lack of integrity monitoring is evident.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 04 Feb 2023
COUNTRY: UAE
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Equipment: Heat exchanger
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

INCIDENT DESCRIPTION:

At 03:38., a fire broke out from the De-ethanizer reboiler during the start-up of Train 3, caused damage to the surrounding equipment structure, piping, electrical and instrumentation cables. Fire and rescue team alerted and extinguished the fire at 06:00.

WHAT WENT WRONG?

Inadequate implementation of QA/QC procedure: During Manufacture, De-ethaniser reboiler channel head partition plate found not fabricated as per the design specification

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Effective implementation of QA/QC during fabrication and installation.
2. To review and ensure all other reboilers conform with design specifications at the next available opportunity.
3. QA to identify and investigate the reasons for any abnormal Observations/indications like imbalanced torquing/ misalignment.

    <<No Barriers Allocated>>

    <<No Causal Factors Allocated>>

NORTH AMERICA ONSHORE

DATE: 16 May 2023
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

While Operators were troubleshooting a well that had tripped, they discovered the gassing gas line of a well had failed in multiple locations and was actively releasing emulsion, casing gas, and steam.

WHAT WENT WRONG?

A combination of steam breakthroughs and linear failure with solid ingress during operation caused additional erosion in the casing gas line, leading to its failure. A previous change in operating philosophy also increased the frequency of steam breakthrough events.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Steam breakthrough management plans on identified at-risk wells should be implemented, including alarms and response strategies. Inflow control devices must properly throttle process to minimize steam breakthroughs.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 02 Sep 2023
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During nightly rounds, the Operator discovered that a 1/2 inch sample line on a pressure vessel bypass line had failed and released sales oil to the building floor and sump.

WHAT WENT WRONG?

Failure of the sample tubing on the vessel due to erosion caused by high velocity process flow. The high velocity of the process within the tubing was due to inadequate port sizes and bend locations.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Tubing port sizes should be adequately sized for their process conditions. Designs and installations should meet company specifications.

BARRIERS:

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 14 Sep 2023
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We walk the line.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During start-up of the main column, operators identified a product release coming from a blind flange. Operators depressurized and isolated the main column.

WHAT WENT WRONG?

A plastic plug had been inadvertently left behind in the bleeder valve from original construction and the plug had failed, causing the release of diluent.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The QA/QC process for verifying item deficiencies should be revised to include further steps and ensure any deficiencies are caught early.

BARRIERS:

Management System Element Barrier Failure: Assurance, review and improvement

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 24 Nov 2023
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A well fire was reported by a member of the public. Upon responding, operations confirmed that a NGL pipeline had failed and was on fire. The line was isolated and the fire self-extinguished when the line de-pressured. Prior to the fire, operations had already been enroute to the site to perform work at the location of the failure, but had been delayed by a phone call. No injuries were sustained.

WHAT WENT WRONG?

Pipeline integrity was compromised by a tether during the inline inspection. Debris observed during the inspection was not initially recognized as a serious threat to pipeline integrity and information about the debris was not communicated to the Operations group prior to start-up. The multiple bends on the pipeline were not accounted for during the work planning and hazard assessment phase of the inline inspection.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Drawings/prints should be up to date for work planning processes. Turnover processes should be formalized to ensure all necessary information is communicated.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 22 Nov 2023
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Equipment: Meter
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Natural gas flow meter failure.

WHAT WENT WRONG?

Tube seal failure, pipeline gas pressure went to transmitter head.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 06 Sep 2023
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Tubing loss of containment on drain line.

WHAT WENT WRONG?

Failure (rupture) of drain line tubing at a bend.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 11 Feb 2023
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Casing Leak.

WHAT WENT WRONG?

Punctured hole in the pump casing.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 21 Apr 2023
COUNTRY: USA
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A roustabout crew was in the process of removing fluid from the slop tank to begin the cleanout process. When they removed a 6-inch blind the crew placed a barrier below the 6-inch nipple to capture the expected fluid release from the tank. After the blind clamp was removed, the fluid overwhelmed the containment releasing fluid to the environment. All released fluid remained inside permeable containment.

WHAT WENT WRONG?

Fluid overwhelmed the containment, releasing fluid to the environment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review proper cleanout procedures.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 28 Mar 2023
COUNTRY: USA
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Other production
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A Contractor was in the process of decommissioning a tank battery when a 4-inch swage connection was removed from the rear of the tank near the manway. The crew was acting on the assumption that the tank was empty but instead the tank contained produced oil. When the connection was removed, the produced fluid was released to environment.

WHAT WENT WRONG?

Crew did not confirm that tank was empty before beginning to remove connections.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review proper clearing and decommissioning procedures with crew.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

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DATE: 06 Jan 2023
COUNTRY: USA
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Well intervention / Well servicing
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A well emulsion line broke during hot oiling operations to remove a line restriction.

WHAT WENT WRONG?

Emulsion line Maximum Allowable Operating Pressure exceeded for hot oil operation and mechanical bending (deflection) of the piping exceeded limits and added additional stress to the piping because it was rubbing against another steel pipe.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Maximum allowable working pressure of piping unknown prior to starting activity.
- No routine PM for emulsion lines in place.
- Improper line installation. Pipeline installation guidelines not followed.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 03 Jan 2023
COUNTRY: USA
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Completions
MODE OF OPERATION: Completion
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

PSE release from WH.

WHAT WENT WRONG?

Bridge of wellbore debris above tubing anchor which led to the spill.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 21 Aug 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A hospital admission and or fatality of a third party

INCIDENT DESCRIPTION:

Oil hauler was pulling a load when the driver noticed the truck caught fire.

WHAT WENT WRONG?

Truck was not grounded properly.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not Available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 11 Jan 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A Facilities Consultant was working near the spill location and noticed fluids exiting the ground at approximately 08:00. He immediately notified the Operations Supervisor and Pipeline Tech. The Pipeline Tech isolated the line. Civil was notified and super sucker and vacuum trucks were dispatched to location.

WHAT WENT WRONG?

Biocide injection was not included in the pipeline treatment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Develop a procedure for analysis and chemical treatment based on best practices.

BARRIERS:

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

DATE: 15 Jul 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Prior to pulling the tubing hanger on the well, the well was equalized over the hanger @2200 psi. The order was given to Contractor employee to back out the hanger pins once the pressure was equalized. Due to the position of the pin under the frac valve, the contractor was unable to see the pin while backing it out. The pin was backed out fully, but the contractor continued to back out the hanger pin and packing gland. This caused the pin and the packing gland to eject from the hanger flange and land approximately 30 feet from the well. The well flowed gas to the atmosphere for 7 minutes before the well was killed.

WHAT WENT WRONG?

JSA did not address gland nut associated risks when backing-out the pin.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Investigate alternatives for future wells tubing head design.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 24 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

A production facility's storage tanks caught fire and failed during startup when the flare backflashed through the tanks' vent header.

WHAT WENT WRONG?

The flare header from the storage tanks had a flammable mixture of natural gas in it from normal atmospheric tank breathing and air ingress from the flare tip into the flare header. The flare flame backflashed into the flare header. There was a detonation arrestor in the flare header but eventually it failed and allowed the flame to reach the storage tanks and they caught fire and failed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Additional barriers/safeguards are needed to prevent a flammable mixture in the flare header.

Startup procedures need to be updated to Identify when a Production Facility needs to be purged with N2 prior to starting it up for operation and include sampling requirements to verify that there is not a flammable mixture in the flare header prior to lighting the flare.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 25 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Completions
MODE OF OPERATION: Completion
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During well frac operations, well to well communication occurred with an adjacent well and its production flow line ruptured and released natural gas and emulsion.

WHAT WENT WRONG?

The adjacent well's wellhead wing valve was not closed and allowed the higher pressure from the well that was being fracked to overpressure the production flowline. The production flowline check valve was incorrectly installed and separated at the threaded connections from the flow line at a pressure lower than its design operating pressure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- The adjacent wells' wellhead wing valves should have been blocked in during the frac operations.
- The engagement of the male and female threads was insufficient to hold the flow line design pressure.
- The threads on the check valve's male and female connections were not compatible.
- No standard instructions existed on how a threaded connection should be made.
- Insufficient training/competency for people cutting/threading pipe/nipples and making threaded pipe connections.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 10 May 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

There was a natural gas release from a buried offsite fuel gas pipeline.

WHAT WENT WRONG?

A previously repaired section of the poly fuel gas pipeline failed.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

A bad fuse was made on that joint of poly pipe when it was repaired previously because the piping was too cold to allow the joint to fuse properly.

The pipeline settled and put additional stress on the piping joint because improper material/procedure was used for backfilling the poly pipeline.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 23 Oct 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During startup of a propane transfer system in a propane storage terminal, a transfer pump's double mechanical seal failed completely, and a pipe flange leaked, releasing propane into the building where the pump was located. The gas cloud migrated out of the building and drifted toward a public road and neighbouring home. City Emergency Response personnel closed the road and evacuated the house until the gas cloud cleared.

WHAT WENT WRONG?

The propane transfer pump's double mechanical seal failed and there was irregular bolt loading of the pipe flange allowing it to leak.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- There were no remotely operated valves available to isolate the leaking equipment to mitigate the size of the propane release.
- There was no level indication or alarms on the pump seal pot.
- Thrust Bearing failure in the electric motor was the most likely cause of catastrophic seal failure.
- Excessive vibration may have led to seal failure.
- Flange bolts may have loosened over time from thermal cycling, over torque or the bolt tightening sequence.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 21 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Power generation unit
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

A fire started within the heater treater/separator building at the tank battery leading to a release of an unknown quantity of fluid. The facility was remotely shut down and the wells feeding the facility were shut in. Emergency services were notified, and the fire and residual heat were put out and suppressed with water. An investigation into the cause of the fire is ongoing.

As the Night Lead was driving up the lease road he noticed an orange glow to the east. The Night Lead called the Select hand. Select hand called Lead back around 2:57 am and said the battery was on fire. Lead told him to park away from the location. The Night lead called the IOC at 2:59 am and ask her to remote ESD the facility and to call the fire department. The Night Lead arrived at the location around 3:05 am parked on the well pad far away from location to the south. The Night Lead and hand hard shut in the three wells to facility. The Night Lead called their supervisor at 3:10 am and notified him what was going on and what was all taken care off. The Fire department showed up around 3:30 am and put the fire out.

WHAT WENT WRONG?

Controls or preventive systems inadequate, No Risk Assessment, HES Procedures or Safe Work Practices inadequate, Design did not anticipate the conditions, Mistake or mental slip.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

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DATE: 26 Apr 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A shut-in tank battery was receiving oil from sales line through the Lease Automated Custody Transfer (LACT) unit back feeding from a faulty check valve, filling up tanks which then caused the LACT unit to kick on. LACT unit went into divert causing a 2-inch tee to rupture due to manual valves being closed going back into tanks. A total of 123 barrels of crude oil spilled to land.

WHAT WENT WRONG?

Isolation valve location allowed for trapped pressure. No procedure for communication with third party LACT operations/repairs. Thermal Expansion hazard not fully recognized.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 16 May 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A release occurred on the flowline due to failure of a 4-inch FlexPipe. The hole where the failure occurred was approximately 3 inches in diameter. 115 MSCF (3300 kg) gas was released to the atmosphere throughout the event.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 16 May 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

An employee was at the well head to get the casing pressure. When he opened the valve, the taped bull plug flew off, causing gas to exit from the valve and hit the employee in the face. There was 170 pounds on the casing when the employee opened the valve. 25.4 MCF (671 kg) gas to air was released in the 3 minutes the valve was open.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

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DATE: 12 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Release of 10 barrels of oil. Skim oil tank was being manually checked every hour but was not checked for one hour and twenty minutes and overflowed in that time.

WHAT WENT WRONG?

Design review failed to uncover inadequacies in design.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Communication between operator and vac truck driver did not occur until after the spill (Contributing Factor). Elimination of Vac truck into Skim Tank. The atmospheric storage tank currently has tank level monitoring system.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

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DATE: 23 Aug 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A production specialist heard the sound of gas being released. Upon investigation, he could see/smell gas blowing out of a section of piping on the bottom side of the 3rd stage bottle of the unit. The production specialist shut down the compressor and waited for gas to dissipate. He found that the flange was threaded and had blown out a nipple valve.

WHAT WENT WRONG?

Design review failed to uncover inadequacies in design.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

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DATE: 13 Nov 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A communication error caused 4 wells to not shutdown, resulting in 3 separators being over-pressurized, releasing 100 barrels of crude, 328 barrels of produced water and 1,036 mscf of gas from separators' PSVs onto facility pad and to adjacent habitat.

WHAT WENT WRONG?

Operations procedures did not exist, Design standards inadequate.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 21 Nov 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We walk the line.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

From information provided by a Lead Field Specialist, a gas-lift compressor blowdown valve was left open to the maintenance tank. It was estimated that the valve was open from approximately 04:15 until 07:15, when it was discovered and closed.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 06 Apr 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Thunderstorms were actively passing through the area at the time, and the employee reported witnessing lightning strike the tank battery. A short time later, flames and smoke became visible. Seven oil tanks were damaged by the fire. No personnel were on site at the time of the incident. There were no reported injuries.

WHAT WENT WRONG?

1. Seven isolated oil tanks had not been cleaned prior to the incident, and vapours from the tank were ignited when struck by lightning.
2. The Abandonment in Place of Out of Service Equipment Policy did not specify a time frame for cleaning isolated tanks to be taken out of service.
3. The buy-back gas line was not equipped with a pressure safety low (PSL) which required manually shutting in the line instead of allowing for a remote shut-in.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Update the Abandonment in Place of Out of Service Equipment Policy to specify a time frame to complete preservation of tanks and other out of service equipment.
2. Evaluate lightning mitigation systems to determine effectiveness and document applicability for installation on sites in proximity to public receptors. Include a requirement to install lightning protection at all sites with proximity to public receptors based on findings.
3. Evaluate remote shut-in capability at buy-back meters and document findings
4. Inventory all similar facilities for lightning mitigation.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 05 Sep 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We walk the line.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

After performing a strainer cleaning and removing the lock out, the valve that is used to verify zero energy for the work area was inadvertently left open. The compressor was then started remotely. This caused a gas release to the atmosphere. No personnel were in the area. There were no reported injuries. Personnel found the open valve when investigating why the compressor failed to start.

WHAT WENT WRONG?

1. The de-isolation of central delivery point CDP compressor #1 was completed but the de-isolation column of the compressor isolation log was not completed. A 3/4-inch vent valve on the discharge of the compressor was left open and gas exited the valve.
2. The vent valve was not one of the valves on the compressor isolation log, and the de-isolation column of the compressor isolation log did not require a verifier.
3. The 3/4-inch vent valve was commonly used to verify zero pressure prior to opening the flange due to its proximity to the work. There was no flagging/tagging present on the valve when found open, and the electronic energy isolation tracker used during normal operations was not in use at the time of the event.
4. Organizational pressure to get the unit online which may have resulted in rushed work leading to the incident.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Revise the Isolation Log to include all valves used for isolation or verification of zero energy.
2. Develop guidance on when to use the energy isolation tracker consistent with normal operations during the commissioning process.
3. Conduct a review of past investigations and verifications to identify Energy Isolation Gaps and develop an implementation plan to close the gaps found during the review.
4. Develop and document a personnel plan to confirm adequate staffing for commissioning and putting online of future facilities.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Policies, standards and objectives

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 18 Dec 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: We walk the line.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Release.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 18 Dec 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas lift line leak.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 11 Dec 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

NGL dump line release.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 08 Dec 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Production facility fire.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 08 Dec 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Tank overflow.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

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DATE: 13 Nov 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We walk the line.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Circulation pump valve leak.

WHAT WENT WRONG?

1" line on circulation pump was open and plug was out.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

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DATE: 09 Nov 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leak.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 11 Oct 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Sales gas line leak due to corroded pipe.

WHAT WENT WRONG?

Internal corrosion.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 19 Sep 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Compressor PRV release.

WHAT WENT WRONG?

Mechanical damage/failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 08 Sep 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Relief, Vent and Discharge Systems: Relief valve (body, plugs)
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

A hospital admission and or fatality of a third party

INCIDENT DESCRIPTION:

Rental compressor Incident.

WHAT WENT WRONG?

Valve cap released due to being under pressure, injuring tech.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

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DATE: 01 Sep 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

MSO received active hi hi alarm at facility. Lacts had been down for monthly proving. MSO called 2nd MSO that was on the route with him to ask him to investigate. 2nd MSO reported that all looked fine but the lact was not running. Original MSO showed back up to find that oil had exited from the tanks and determined that the new ZZ battery was still producing into the facility and sales lact was still down. Dimensions of spill were (115ft x 54ft x 2in) resulting in a total release of 19.3442 bbls, all of the spill was oil. 19 bbls were recovered. The cause of the leak was failure of the lact to restart. 2nd MSO had last been by the area approximately 3 hours previously. The release was contained on pad.

WHAT WENT WRONG?

Failure of the lact to restart.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

DATE: 02 Aug 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Compressor fire.

WHAT WENT WRONG?

Valve cap suspected failure causing a gas release and fire.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 11 Jul 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Production header flange release.

WHAT WENT WRONG?

Pinhole release.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 11 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Crude released to secondary containment due to exterior corrosion.

WHAT WENT WRONG?

Pipe was severely corroded on the outside.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 10 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We walk the line.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Ball valve on top of pump became open due to vibration, causing release.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 01 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Vac truck driver unloading unaware of tank level.

WHAT WENT WRONG?

Competency, incorrect isolation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 20 Apr 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Planned shutdown
POINT OF RELEASE: Equipment: Pressure vessel
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Nipple on belly came off while replacing a KO causing release.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 14 Apr 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Spill due to pinhole in tank.

WHAT WENT WRONG?

Internal corrosion made hole at bottom of tank causing leakage.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 24 Feb 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 1
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Worker was managing high fluid levels in the low pressure knock out (LPKO) vessel while bringing wells online at the facility and was found unresponsive.

WHAT WENT WRONG?

Oxygen depleted atmosphere. Unexpected release.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review small building design with focus on ventilation and alarms. Reinforce existing practice of not draining process fluids within enclosed spaces or near ignition sources. Reinforce use of personal multi-gas protection monitors to verify acceptable atmospheric conditions, especially prior to building / enclosure entry and during activities involving process fluids. Ensure lone worker policies and safety-critical tools are functioning as intended – including calibration and testing – and supervisors are accountable for monitoring use and adherence.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

DATE: 07 Feb 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil Spill – valve failure due to weather.

WHAT WENT WRONG?

Weather froze valve and broke.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 21 Jan 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Lease Operator responded to facility alarm and found nipple connection leaking produced oil. Facility was shut down and Supervisor notified.

WHAT WENT WRONG?

Nipple failed due to vibration.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review site inspection and maintenance processes.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 30 Jan 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 09:30, mechanic foreman was notified of a gas leak on the gas lift recycle valve. Foreman dispatched a mechanic to the location, unit was isolated and the mechanic found a cracked nipple. Necessary repairs were made. Foreman and SSHE were notified. No waters were affected.

WHAT WENT WRONG?

Nipple crack caused by vibration.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review site inspection and maintenance processes.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 11 Feb 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Driver loading truck left the loading control area and was not monitoring the loading operation. Overfilled the tanker trailer.

WHAT WENT WRONG?

Driver inattention.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review site inspection and maintenance processes.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

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DATE: 10 Feb 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

A Production Tech and a maintenance crew were prepping for a job at the central tank battery when they heard a loud noise and saw flames coming from two fiberglass tanks at the adjacent SWD facility. All personnel immediately left the area. When they reached a safe distance, the fire department and supervisors were notified. When the fire department arrived, they applied water and foam to the tanks and surrounding area. No injuries were reported. A total of 50 barrels of produced water, fresh water, and foam were recovered.

WHAT WENT WRONG?

The specific failure was not identified, however an equipment failure causing sparking potential, grounding failure, or anti-static protection is the most likely cause of this fire.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Identify cost effective options for tank and containment liner replacement. Ensure that replacement equipment has appropriate grounding and anti-static protection.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

DATE: 13 Feb 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Equipment: Pressure vessel
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A lease operator reported a release of oil at the central tank battery. Crew on site was in process of cleaning bulk production separator and began emptying all fluid from it. The crew believed that the separator was empty, and they opened the manway cover to inspect the splash plate. Opening the cover resulted in 7.78 barrels of oil spilling onto the pad surface.

WHAT WENT WRONG?

Crew did not verify that separator was empty before opening manway cover.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review equipment clearing procedures and training.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

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DATE: 14 Feb 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At approximately 07:30, Production Tech received a call from a neighbour reporting a fire on the location and the local fire department had been dispatched. The PT notified the Production Foreman and Area Foreman. PF and PT travelled to the location and observed two tanks had been destroyed by fire from an apparent lightning strike and shut the well in. The Fire department had just finished leaving the location after discovering the fire was already extinguished and spraying precautionary foam.

WHAT WENT WRONG?

Suspected lightning strike.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review site inspection and maintenance processes.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

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DATE: 05 Feb 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At approximately 05:45, a ½-inch swaged up to 1-inch fitting broke off of the high-pressure lean glycol going to the heat exchanger on the glycol contactor tower due to a lack of bracing for about 15 feet of 1-inch piping. A production lease operator noticed the leak and notified group on-call. On-call mechanic parked his truck at the entrance to the facility due to natural gas leaking out of the ½-inch fitting.

WHAT WENT WRONG?

Lack of bracing.

Design/installation less than adequate.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Create process to track progress of events for which metered data is not available – supplement to existing process for flaring incidents.
2. Install bracing to support piping and pump.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 04 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 10:00, a lease operator was performing their rounds when the operator discovered that oil had been released from the recycle line after a cap to the recycle pumps screen came loose. The lease operator isolated the lines and re-installed the cap on the screen.

WHAT WENT WRONG?

A bolt was missing on the pot's strainer, allowing for the oil to be spilled into lined containment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance practices.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 06 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Abandonment
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A landowner was dumping dirt from a pull-behind scraper and slid into a wellhead breaking one of the wellhead valves and causing a gas release and spill. Equipment operator drove away from the site and notified Production Tech. Notifications were made to Leadership and SSHE. PT, Foreman, and Area Foreman arrived on scene, 16:33. Area foreman confirmed well dead. Notification made to Field inspector via phone call, left message. Vacuum truck was dispatched to recover free standing fluid. Repairs have been made and some absorbent spread, additional removal and remediation are being scheduled. No Waters were affected.

WHAT WENT WRONG?

Mobile equipment strike.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance practices.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

DATE: 05 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pressure vessel
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At around 23:45, operations were notified by contract water hauler about a fluid release out of the heater treater due to a seal failure on top of the water leg. Flow hand was notified by foreman of the leak and arrived at the location and isolated the leak and shut in the well.

WHAT WENT WRONG?

Material failure of seal.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance practices.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 16 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Load line valve at tank parted allowing a release of 30 barrels of produced water and 90 barrels of condensate into permeable containment.

WHAT WENT WRONG?

Valve failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Recommended to remove 2-piece valves from service.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 14 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

Contract lease operator was performing work in the enclosed dehydration unit when glycol (est. 360-375 Fahrenheit) was released from the unit contacting the IPs body. A contract automations tech. stopped to assist with first aid and provided transport to ER. The IP was given medical treatment beyond first aid and was admitted to the burn unit for evaluation and treatment.

WHAT WENT WRONG?

Under litigation hold.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Under litigation hold.

BARRIERS:

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: 18 Mar 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At approximately15:45, a plant operator reported a gas release event. Two plant employees entered the building, sourced the release point, and were able to stop it. No injuries occurred.

WHAT WENT WRONG?

Economizer valve failed allowing gas to leak by and be released.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance practices.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 13 Apr 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At approximately 12:30, a plant operator received a LEL alarm (20%) from the compression building. The operator turned the building ventilation system on to displace the LEL's. Once the LELs had dissipated and operator cleared the building and entered to ESD the compressors. After the compressors had been ESDed and no LELs were present in the building operations began to source the leak. It was found that one compressor was leaking gas. The unit was blown down and isolated from the process for repair.

WHAT WENT WRONG?

Equipment failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review maintenance and inspection procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 26 Apr 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Completions
MODE OF OPERATION: Completion
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Frac Blender caught fire during frac operation and spread to two pump trucks before being extinguished by local fire department.

WHAT WENT WRONG?

Clutch bearing failed causing clutch plate to spin out of its housing, striking an adjacent hydraulic pump which released hydraulic oil and ignited.

Blender Operator did not activate blender e-kill switches prior to evacuating their post on the blender.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Add fire suppression verbiage to Frac WOSPs. Blenders and frac pumps are required to have automatic fire suppression systems installed and operable prior to fracking. In addition, one portable fire suppression unit will be kept onsite and confirmed operable prior to fracking.
2. Evaluate location of e-kill switches and move, as deemed appropriate, to locations on equipment away from where personnel hazards may occur or replace with switches that automatically kill the equipment (ex. 2. vibration switches).
3. Evaluate capabilities of service party fire suppression support and determine if continued use reduces risk.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Response to emergencies

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 26 Apr 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Plant Operator noticed steam coming from the Glycol Re-Boiler area during heavy thunderstorm and hail event. Operator notified Supervisor and began shutting system down.

WHAT WENT WRONG?

It was found that a thermocouple for a temperature indicator had developed a leak, allowing heat-medium to release to impermeable secondary containment and atmosphere.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review plant inspection and maintenance procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 01 May 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

At 17:23, a contractor crew was preparing for a pipeline tie-in. While performing a bevel cut with a torch to remove a section of pipe, a flash fire and liquid condensate fire occurred resulting in burns to three contractors.

WHAT WENT WRONG?

Under litigation hold.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Under litigation hold.

BARRIERS:

Unspecified: Unspecified

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: 15 May 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Well intervention / Well servicing
POINT OF RELEASE: Wells, drilling and intervention: Mud circuit/tanks
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

While circulating bottoms up prior to washing over a fish, the well returned gas cut fluids to the tank. The pump operator noticed the gas and engaged the emergency shut down and instant neutral simultaneously, but the engine began to run away and ignited the remaining gas in the area. The well was secured via hydraulic closing unit and TIW valve, and site evacuated. Emergency services were called and the local fire department arrived on scene to extinguish the fire.

WHAT WENT WRONG?

Gas reached engine and caused runaway.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

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BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 03 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Foreman was notified of a lightning strike and fire at a common point tank battery. No one was on location or in the area. The fire burnt out on its own and released fluids were contained inside secondary containment. All six 400 bbl steel tanks were damaged (3 of the tanks were out-of-service).

WHAT WENT WRONG?

Lightning strike.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

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BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

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DATE: 04 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 03:00, a flowback hand noticed that the tank battery was overflowing oil into the impermeable containment. The well was isolated, and vacuum trucks were called to recover the free-standing liquid out of the containment.

WHAT WENT WRONG?

A PW load was pulled at 15:00 on the same day and at this time the well was making 8 barrels of Produced Water (PW)/hr. Over the next 12 hours the well started making 41 barrels PW/hr causing the tank to overflow due to unforeseen increases in production.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Improve risk assessment for abnormal conditions / production amounts.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 08 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At approximately 11:45 CT, a plant operator was on the board when the operator noticed elevated levels in the compression area (approximately 9% LEL). The operator notified the onsite supervisor, who turned on the buildings ventilation to control the LEL levels in the building. The operators were able to remotely shutdown the units with the help from the automation supervisor. The plant operators allowed for the LELs to dissipate from the building prior to allowing anyone into the building to source the leak.

WHAT WENT WRONG?

While sourcing the leak, plant mechanics discovered a cracked weld on the 1 stage discharge pulsation bottle.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review maintenance and inspection procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 08 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The lease operator reported the venting hatches on both skim tanks lifted due to the water dump washing out on the bulk separator. This caused oil and gas to exit the hatch and release into impermeable containment.

WHAT WENT WRONG?

Washout.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Evaluate dump valve, piping, gun barrel and PW levels (Hi/HiHi alarms) to address PW spills on dump valve failures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 09 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

While opening a well to production after well work handed the location back to production, equipment failed, allowing a LOPC from the casing valve to atmosphere.

WHAT WENT WRONG?

A LOPC occurred after an overpressure event and check valve failure. The flowline and gas lift line parted from the well head, causing a well control event.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Communicate hazards to field practitioners relating to pressure surge and ensure pressure is raised in systems gradually.
2. Conduct an evaluation to determine which check valves are fit for service.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 17 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 21:39, an electrical storm caused a malfunction in the level controller for the inlet slug catcher at the facility allowing gas to flow to the gun barrel and over pressurize the system releasing gas, condensate, and produced water to the environment.

WHAT WENT WRONG?

Lightning caused malfunction in level controller.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

DATE: 25 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 15:45, 12 barrels of oil were released through the thief hatch on top of the skim tank during battery startup operations. 3 barrels was recovered with a vac truck. Spill was contained in tank berm/wall and did not leave location.

WHAT WENT WRONG?

Valve malfunction.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 30 Jun 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 03:14 CST, a 3rd party company was in the process of excavating their right of way (ROW) to expose a produced water line that had been leaking, when they struck an un-marked 4-inch poly gas line with the bucket of the excavator.

WHAT WENT WRONG?

3rd party submitted One Call ticket and the line that was struck was not marked. Area in One Call ticket may not have matched work being done.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Authorization of temporary and mobile equipment

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 01 Jul 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Night pumper received a LEL alarm for the compressor building. At 23:53 the night pumper received a HiHi LEL SD alarm for the compressor. Upon inspection of the building, it was determined that an LEL release had occurred due to a cracked fitting on the 3rd stage compressor suction piping.

WHAT WENT WRONG?

Cracked Fitting.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 27 Jul 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The production Tech received a level high alarm on vessel, while investigating cause of alarm he discovered the skim tank releasing fluid to the environment through the thief hatch.

WHAT WENT WRONG?

Level alarm failed to sense correct level of foaming liquid.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Install Guided wave radar.
- Install longer tuning fork.
- Install dump metering.
- Lower vessel pressures to decrease gas content of liquid stream.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 30 Jul 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Plant Operator was on the board when the operator noticed elevated levels on the building detectors in the compression area. It was discovered that the 1st stage discharge pressure transmitter ¼-inch stainless line on the discharge bottle was the source of the leak.

WHAT WENT WRONG?

Equipment Failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review maintenance and inspection procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 23 Aug 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A 3-inch poly flowline was found leaking in the area while making routine rounds. The poly line failed in a weak spot when the well was turned back on after being down for several months causing a release to the environment.

WHAT WENT WRONG?

Material failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 21 Sep 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Lease Operator discovered spill, called for assistance to perform isolations. Flowline found with failure resulting spill.

WHAT WENT WRONG?

Material Failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance process.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 04 Oct 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Other production
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Contract bush hog operator cut between header and chemical tank before area was cleared by weed-eater crew. Frame of bush hog hooked chemical injection line and broke off valve at injection point.

WHAT WENT WRONG?

Unseen, unburied chemical injection line due to lack of location maintenance and bush hog operator working before area was cleared by weed eaters.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

No personnel should be within 100' of active bush hogging operations, bush hog operator should not enter areas that have not been previously cleared for cutting.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

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DATE: 03 Oct 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Lease operator arrived and found product releasing from thief hatches (three saltwater tanks) spilling into permeable secondary containment. Transfer pump shut down on high line pressure results in water stacking in tanks leading to spill.

WHAT WENT WRONG?

The only path to overflow tanks was through a single production tank creating an error prone situation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Address overflow isolation potential.

Explore standardized SWD pump control narrative.

Improve hand shift change effectiveness.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 22 Oct 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A HIHI tank level alarm was received along with a report from an operator that the tank was already overflowing into impermeable containment.

WHAT WENT WRONG?

Failure to respond properly to high level alarm.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review alarm response procedures with operations.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 31 Oct 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

On 31 October 2023, at around 04:00, the gas lift compressor at the facility experienced a high-high alarm, leading to a shutdown. Leak detection and repair team responded on 6 November 2023, and identified a leak from the compressor's 3rd stage temperature probe.

WHAT WENT WRONG?

A temperature gauge was installed onto the compressor skid without a thermo-weld, causing the gauge to break, releasing gas into the building.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review installation and inspection procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 09 Nov 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Mechanic arrived on location to start the compressor after it went down on 3rd stage liquid level. The mechanic heard gas leaking from the 3rd stage cylinder while the compressor was down.

WHAT WENT WRONG?

Equipment Failure – Wear and Tear.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 06 Dec 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A contractor was digging a hole to set a power pole and contacted a buried flex steel line.

WHAT WENT WRONG?

The line had not been identified on the one call ticket or the contractor's secondary sweep.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review sweep procedure.

BARRIERS:

Human Barrier Failures: Authorization of temporary and mobile equipment

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

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DATE: 13 Dec 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Heat exchanger
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Gas lift unit shut down on High-High LEL alarm after a crack developed in the coolers' 3rd stage discharge line.

WHAT WENT WRONG?

Material Failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review inspection and maintenance processes.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 21 Dec 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Operations received a phone call from Measurement advising there was a discrepancy in the field volume and total sales. Operations immediately mobilized to investigate and found a section of pipeline that was losing pressure when the valves were shut. They started backtracking and found a section of pipeline that was leaking.

WHAT WENT WRONG?

Two holes were discovered in an 8-inch gas pipe line.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Review pipeline inspection procedures.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

NORTH AMERICA OFFSHORE

DATE: 06 Jan 2023
COUNTRY: Canada
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Sample system
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

While attempting to collect a sample from the Subsea Hydraulic Power Unit (HPU) supply tank, the mechanic found the end cap was tight and hard to remove. After applying additional force with their left hand, the cap loosened and released high-pressure hydraulic fluid, contacting the mechanic's left index finger, and resulting in an injection type injury.

WHAT WENT WRONG?

Mechanic took sample from wrong location (HP side of Subsea Hydraulic Power Unit (HPU) rather than LP side).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Sample points should be clearly labelled to avoid confusion.

BARRIERS:

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 27 Feb 2023
COUNTRY: Kazakhstan
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Meter
PROCESS SAFETY FUNDAMENTAL: We control ignition sources.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

A gas leak from a multi-phase flow meter was detected by an onsite H2S detector at a well. The Field Section Supervisor and Senior Operator received a notification and arrived at the location with two Company workers. The system was isolated, and residual pressure was being released from the leak point. After 30 minutes, the two Company workers began removing the insulation from the pipe to identify the leak point. They were wearing self-contained breathing apparatuses (SCBAs) and flame-resistant clothing. An ignition occurred as the insulation was being removed. Both workers safely evacuated with no injuries. The fire was extinguished by ERT with no further injuries.

WHAT WENT WRONG?

Hazard not recognized, Safeguards that Did Not Work, Inadequate written communication.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Multi-Phase Flow Meters utilized soft insulating material that absorbed hydrocarbons, which contributed to intensive combustion (other materials have better properties against combustion).
2. There is no written procedure for access to the multi-phase flow meter stations.
3. Leak test was carried out at 90 bar but some wells may have exceeded operating pressure parameters.

BARRIERS:

Hardware Barrier Failures: Ignition Control

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

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DATE: 06 May 2023
COUNTRY: Turkmenistan
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Well intervention / Well servicing
POINT OF RELEASE: Wells, drilling and intervention: Mud circuit/tanks
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

At around 03:45, an accident occurred during the milling operations of a bridge plug in the Workover. During the circulation a kick of gas occurred, which probably ignited the fire on the pumping area. The crew immediately mustered in the safe location. The fire did spread near the return tank. No injured personnel. Emergency Response team (Fire team and Doctor) was on the incident site in few minutes ready to take actions. Fire Fighting teams were on site working to kill the fire. All of the affected area was cordoned off and Safety Signs installed in order to prevent any unauthorized access.

Involved the Governmental agency for Well Control which created the line to inject brine in the well.

After roughly 3 hours of injection, the fire got extinguished at 16.45 pm and the well stopped to blow hydrocarbons.

WHAT WENT WRONG?

- Reduced team presence due to Sick Leave management.
- Safety Device: Well Control equipment not fully compliant with Company minimum requirements (BOP, Killing / Choke line, Burning).
- Failure to use operating instruction. Reverse circulation not mentioned in the operative programme.

- Ineffective supervision.
- Underestimation of risk / overestimation of abilities / not respecting the rules, Human Factor and Complacency.
- Stop Work Authority not applied.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Improve well control equipment assessment to reduce the gaps in between actual working status and minimum international standards.
- Equipment checks, readiness and work organization compliance prior job execution.
- Ensure all personnel involved in the operation are aware of the operating instructions and methods of statement (including emergency).
- Any change or deviation to design, asset, process, work procedure or practices shall be accurately identified , reported and assessed by competent and trained personnel.
- Improve awareness on Process Safety Fundamentals.
- Ensure no gaps in competence, awareness and training of the personnel involved in well operations.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Ignition Control

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

RUSSIA & CENTRAL ASIA OFFSHORE

No Tier 1 PSE reported.

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 23 Sep 2023
COUNTRY: Argentina
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Sludge flow through the side valve of the head in an uncontrolled manner, while the insulation pipe packoff was being installed. It is a High Potential Event. One of the side valves in section B of the wellhead must be open during the lowering of the packer, according to the procedure of the supplier company. The well "recovers" energy due to a degradation of the fluid barrier (loss of level in the well), which produces an uncontrolled upwelling for approximately 1 hour. It is possible to close the well in a "manual" way by operating the flywheel of this valve. Final state of packer out of position. The packing procedure should be revised to take into account situations of unstable well and degradation of the primary barrier (sludge).

WHAT WENT WRONG?

Improper development of standards/procedures/instructions.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Update technical specification of heads: Include in the specification compliance with double barrier in washing and packaging tasks.
- Prepare and approve contingency scenario "loss of containment in section B valve" with tower equipment companies.
- Protocol design and implementation of pilot test of valve with remote surface actuation.
- Standardization of good practices: Update MPD Specification: Detection and Cure of Safe Intake Zones.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 12 Jan 2023
COUNTRY: Argentina
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We walk the line.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Casing 5" csg, at the time of filling the well at 4171 meters, fluid was observed in the cellar and it was dispersed with 2 cubic metres, affecting a 54 square metres of soil, containing it. We proceed to suck the cellar with Contractor Company personnel and check the section B valve on the n-o side, which was missing 2 ½" to close it with the presence of the team leader, shift manager, PM from Contractor Company and CMR. It was pumped again directly, observing that the mud returns through the screen and said valve did not leak.

WHAT WENT WRONG?

Spill/release/uncontrolled discharge to the environment.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 29 Mar 2023
COUNTRY: Argentina
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Mud circuit/tanks
PROCESS SAFETY FUNDAMENTAL: We walk the line.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Drilling with the MPD system shows a pressure increase of 500 psi as a result of the flow. HCR was opened according to working diagram. When HCR was opened, an increase in leakage pressure was observed in remote shock. When the fluid leaked, it was directed to the equipment picker and because the discharge line valve was closed, the fluid was directed through the vent line to the burn pit in which the fluid was contained.

WHAT WENT WRONG?

Inadequate communication between work teams.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

A meeting was held with staff on duty to remind them of the need to take into account the alignments of the manifold.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 13 Oct 2023
COUNTRY: Argentina
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Carrying out tool weight taking at 5530 meters, the 6" hose (low pressure 150 PSI) return to MPD striker breakage, generating a spray/spill of approximately 3000 litres of mud affecting 16m2 of location land.

WHAT WENT WRONG?

Difficulty with equipment/design/design specifications/design does not follow specifications.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Install hoses according to the design established in the Layout (6" WP 175 psi AISI 316SS internal and AISI 304SS external) or equivalent.

Evaluate the possibility with engineering of installing 6" rigid line from the outlet of the Coriolis to the floor of the location and entrance to the separator.

Send the remains of cut hoses to supplier so that they can make an analysis of why this event may have occurred.

Install 2" relief valve in rigid line in flowline according to the Layout with discharge to the thumper.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 31 Dec 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Choke
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The surveyor reported that the well pad cpo cover had broken.

WHAT WENT WRONG?

Equipment Failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 02 Feb 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Sump/pit overflow
PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

PCI dispatch pumps had stopped, the operator repeatedly tried to put them into service, not responding to the command to start any of the pumps, either locally or remotely. As a consequence, TK 53 of PCI LC overflow was generated, remaining contained within its enclosure.

WHAT WENT WRONG?

Spill/release/discharge to contained environment – equipment failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 07 Feb 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

An alert was received for a spill in an oil pipeline on the main road, approximately 50 metres from the well.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 03 Feb 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During normal operation, PSV caused leakage through 10" flange gasket from the emergency line valve discharge flange. Affected area: approximately 300 square metres. Volume: approximately 4 cubic metres.

WHAT WENT WRONG?

Spill/release/discharge to the environment not contained.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 11 Feb 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Condensate spillage occurred by overflow chamber.

WHAT WENT WRONG?

- Failure to verify/monitor/observe/analyze
- Failure to comply with policy/procedure/instructions
- Inadequate communication infrastructure/process
- Lack of Skill
- Inadequate review of instructions
- Improper download/deployment of policies, procedures, practices, or guidelines for action

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Have a checklist of the points to be controlled and define the schedules in which the survey and data verification must be carried out. This point must be reflected by means of a registry.

Replicate these improvement actions in all of the facilities and incorporate them into the Operation Programming PPP.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 23 Feb 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

Carrying out mechanical excavation work with a trencher in the progressive 10+00 of the pipeline under construction, the lateral touch of the trencher bar with the 24" duct occurred, causing a gas leak due to damage to the pipe. The trencher operator stopped the equipment and they drove away from the site. No personal injuries were recorded.

WHAT WENT WRONG?

- Inadequate discipline.
- Poor instruction/guidance and/or preparation.
- Inadequate scheduling or planning of work.
- Inadequate communication between work teams.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Establish a follow-up guide for new personnel joining the company or changing position, with roles and responsibilities in the programming process and key procedures
- Implement a typical interference detection report with standardized references, supports to include and minimum requirements in case of subcontracting the service.
- Improvement of the mechanical excavation release act, condition of freehand signature at the end of the tour.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 09 Mar 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Sump/pit overflow
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Spillage of base oil sludge occurred due to overflow of the sieve pool. Approximate affected area of 120 square metres. A spilled volume of 9 cubic metres was calculated, of which 3 cubic metres were recovered. The level of overflow of the sieve pool was exceeded.

WHAT WENT WRONG?

Not available.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 12 Mar 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Plant supervisor reported a fire in the KOD sector. For this reason, resources were activated. It was confirmed that the fire was in the flare sector. Assistance was required because one of the workers was in a state of shock.

WHAT WENT WRONG?

- Engineering/Inadequate Assessment of Operational Conditions
- Inadequate Engineering/Design Development
- Engineering – Inadequate Change Management
- Procurement – Inadequate Selection of Material/Equipment
- Contracting – Inadequate Identification of Needs
- Inadequate Communication of CMASS Information
- Standards – Inadequate Monitoring of Compliance with Standards
- Leadership – Inadequate Long-Term Planning/Poor Instruction

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Generate Agile team.
- Life cycle analysis of chemicals from perfo, frac, production, looking for the cause of impurities that reach the plant and actions to be taken.
- Note: systemic situation seen in other incidents or operational reports/Review design, looking to use direct measurement.
- Apply result in PTC BS, LACh, and future in project, plus PTC Stds designs.
- Review complete KOD system design, including dimensions.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 26 Dec 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the pipeline tour, a spill was detected in the blocking chamber, valve number 1 progressive 10.

WHAT WENT WRONG?

Operational failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 25 Apr 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Piping/valve (inadvertently left) open to atmosphere
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Runner reports spill, activates role. On location, the protocol is activated to close a leaking valve. It can be closed, you can see the C section valve of the well open, apparently due to marks in the hold the plug came out abac. A section C valve is found, from the branch where the transmitter was NOT, therefore the valve should have been closed, which is why it is observed that the plug or abac has been detached on the cellar wall.

WHAT WENT WRONG?

- Failure to comply with policy/procedure/instructions
- Failure to verify/monitor/observation

- Improperly fill out/file/keep records
- Improper review of instructions
- Inadequate communication between work teams
- Inadequate discipline

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Place in the Pulling Programme to reference the protocol points in the programme's maneuvers. Disseminate so that it is complied with in the rest of the assets.
- Prepare a document of the well assurance process, validated by the active and EO managements. It includes reviewing the assurance protocol and including new practices.
- Disseminate the assurance protocol, with signature and registration, to all those involved in the assurance process.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 18 Jul 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Plant operator goes to the crude oil dispatch pump sector in order to put the pump into service since the tank level was increasing, an oil spill was detected in the pump sector due to overflowing of the containment area and drainage chamber. When checking the condition of the pump, it is found to be leaking due to a broken packing. A broken packing press is also seen.

WHAT WENT WRONG?

- Operating Out-of-Spec Equipment
- Improper Usage Planning
- Poor Maintenance
- Failure to Warn/Intervene
- Improperly Filing/Keeping Records
- Inadequate Guards or Barriers
- Inadequate Infrastructure
- Loss of Situational Attention

- Inadequate Communication at Shift Change
- Inadequate Components or Parts
- Inadequate Assessment of Operating Conditions

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Establish between supervision and frequency in cleaning pump suction filters based on the characteristics of the fluid associated with the start-up of new wells.
- Perform in maintenance sector diffusion of the deviation (poor adjustment) to avoid recurrence of failure.
- Establish plan to ensure operational route of installation, to detect deviations early in the installation process.
- Perform MDC to adapt security level detection system.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 24 Jul 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the desanding process of the well, the nipple + safety valve assembly detached and subsequently fell from the upper connection of the sand trap. Due to the pressure, these pieces were thrown approximately 12.9m from the sand trap. As a result of this situation, decompression of the system occurs, generating a gas emission into the atmosphere. There is no damage to people or other nearby facilities in service.

WHAT WENT WRONG?

Not availalble.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Perform hazard and risk assessment related to the design of equipment (pressure vessels) under DTM conditions. Evaluate external forces in this condition at vulnerable connection points (threaded connections).
- Update of assembly and operation procedures, making explicit the removal of the PSV in loading and unloading manoeuvres of the Desander. Update procedures and planning with actions derived from the analysis of hazards and risks related to the design.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 16 Sep 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Sump/pit overflow
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The daytime shift operator enters the plant and on his first tour, finds the Tank Skimmer enclosure with fluid, volume approximately 75 cubic metres inside the enclosure.

WHAT WENT WRONG?

- Operating Out-of-Specification Equipment
- Failure to Identify Hazards and Risk Analysis
- Defective Tool, Equipment, Material or Software
- Equipment or Tool Failure
- Inadequate or Unused Data/Information
- Inadequate Design Development
- Inadequate Design Review
- Inadequate Reference Documents, Instructions and Advisory Publications Available

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Establish processes to ensure that projects deliver to the contractor building the facility all CAO drawings for equipment not purchased/built by that contractor. Include in the contract the adaptation of the ID with CAO of this equipment.
- Establish in the pre-start-up review (In ANNEX II STD ITEM 3.4 and ITEM 3.5, linked to the operability of the control loops) that all the control loops of the installation must be available. Otherwise, it must be discarded.

BARRIERS:

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 21 Nov 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Fired heater/Boiler/Furnace
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

In normal oven operation, for reasons that will be investigated, a fire occurs in its expansion vessel, apparently due to a thermal oil leak.

WHAT WENT WRONG?

Control system failure and operational errors

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 15 Nov 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Plant operator observes leak in the packing of a pump while starting up, generating hydrocarbon leakage through a 3/4 purge valve in the suction line.

WHAT WENT WRONG?

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CORRECTIVE ACTIONS & RECOMMENDATIONS:

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BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 09 Oct 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Equipment: Fired heater/Boiler/Furnace
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

Fire/Explosion damage >$100,000 direct cost to the company

INCIDENT DESCRIPTION:

After corrective maintenance due to problems in the gas regulator, the equipment remains in service without any anomaly being reported and after 3 hours of being in service, a deflagration occurs with openings of the inspection covers.

WHAT WENT WRONG?

Failure due to equipment malfunction

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Ignition Control

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 09 Oct 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

While excavations were being carried out with a backhoe to mount a pole for a low-voltage electrical line, the ERFV conduction line of the field collector occurred, causing a minor spill.

WHAT WENT WRONG?

Operational failure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 11 Sep 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 21:00, Supervisor activates EMERGENCY ROLE to the control room in relation to the spill contained in the parapet of the condensate tank. As preliminary observations, a radar measurement reliability problem is seen and considerable fluid contributions from CABO (bleed water) to the Cutter Tank, which causes the excessive increase of liquids and the reference spill. Affected surface area 238m², determining an estimated volume of 8m3 contained within the parapet.

WHAT WENT WRONG?

Operational failure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

DATE: 07 Sep 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A crude oil spill occurs due to a break in the well conduction pipe. The fluid drains through a slope to the internal road of the reservoir.

WHAT WENT WRONG?

Equipment failure/inadequate protection

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 31 Aug 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Equipment: Filter
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

In manoeuvres for the start-up of a well, working on the different lines that contribute to the 32" collector within PIPH, a sudden increase in pressure was observed in the pumping system, which generated a rupture of the O-ring of the filter and subsequently a stop of pumping towards the Pumping Head.

WHAT WENT WRONG?

Equipment failure/inadequate protection.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 23 Aug 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The loss of hydrocarbon occurs due to overflow of the general tank of the battery after a power outage occurred in the area, the pumps do not restore their service, causing the tank level to increase and a subsequent spill.

WHAT WENT WRONG?

Failure in control system and response after shutdown due to power outage

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 19 May 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

A 24" diameter tank outlet pipe breaks. Call role is activated, tank 3 outlet valves (dispatch) are immediately blocked, recovery tasks begin with vacuum and sanitation units with crews and road equipment.

Spilled volume 19.9 cubic metres of crude oil in specification and approximately 14 cubic metres were recovered. The affected area was 2,032 square metres, affecting part of the installation, access road and surrounding location. Tasks and manoeuvres are carried out to bypass the affected pipe for subsequent repair.

WHAT WENT WRONG?

Equipment/material failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 27 Apr 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Unspecified production
POINT OF RELEASE: Unknown/insufficient information: Unknown/insufficient information
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The present spill, along with 13 others that occurred at the same time in the area, were a consequence of sabotage at company facilities carried out by unidentified people to date. The sabotage consisted of valve actuation, and a surge shaft cover was also removed. Corresponding criminal complaints are being filed in order to identify and hold the perpetrators responsible.

WHAT WENT WRONG?

Sabotage.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 12 Apr 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Platform/Well Pad Flowline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

Two Production Supervisors at a well, detected behavioural anomalies of the Plunger Lift system.

They verified in the field by visual and auditory inspection that the piston arrived dry on the surface. This being an unsafe condition for the integrity of the extractive system lubricator, it was decided to close the well with double blocking, depressurize the lubricator section and check the condition of the piston and anchor device (catcher). The lubricator cover was removed and it was observed that the piston was inverted and that the catcher was broken. Unable to remove the piston, one of the Supervisors proceeds to slightly pressurize it with a manoeuvre valve, forcing the displacement and subsequent fall of the piston. This causes it to impact the right hand of the supervisor who was manoeuvring the reference valve.

WHAT WENT WRONG?

Equipment/material failure and risk identification.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 24 Apr 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During well start-up manoeuvres, the production bridge nipple breaks (Direct), causing well decompression and fluid spillage, affecting an approximate surface area of 50 square metres and an estimated volume of 1.5 cubic metres of oil. During the event, the operator was not at the well location.

WHAT WENT WRONG?

Material Failure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

N/A

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 13 Feb 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The oil pipeline broke, causing a crude oil spill, affecting the soil and alluvial channel.

WHAT WENT WRONG?

Equipment/material failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 11 Jan 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Well intervention equipment
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

In stimulation operation with pumping equipment in well (Injector), the location is entered, a closed well is verified with 190 Kg/cm2 in direct and 0 Kg/cm2 in annular, a Safety meeting was held and AST was prepared, equipment was assembled to pump directly, exhaust lines were connected directly and between pipes, lines were tested at 3700 psi, 2,000 litres of preflow + additives are injected, 2,450 litres of treatment (chlorine oxide) were pumped when a fluid leak occurred through a nipple (fissure) located on the master valve, affecting two employees of the Company that operated.

WHAT WENT WRONG?

Equipment/material failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Equipment/material failure.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 27 Apr 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the Plant tour, a loss was detected in the process drive line, 10" exterior coated steel pipe.

WHAT WENT WRONG?

Quality of repairs.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 24 Feb 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

In the event of a general reservoir power outage, an Emergency Pool spill is detected.

WHAT WENT WRONG?

Inadequate evaluation of operational conditions.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Human Barrier Failures: Authorization of temporary and mobile equipment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 18 Apr 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Pipeline Shuttle Dielectric Joint Blowing Loss

WHAT WENT WRONG?

-

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

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DATE: 27 Apr 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Environmental incidents that occurred due to the breakage of a 10" steel pipeline.

WHAT WENT WRONG?

- Equipment/tool failure
- Inadequate guards or barriers

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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ZDATE: 12 May 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Loss due to disengagement of production bridge in 2" threaded ball valve body opening.

WHAT WENT WRONG?

Poor inspection and/or control/

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 21 Jul 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Compressor/blower/fan
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During an operator's tour, a leaking crude oil pump was found.

WHAT WENT WRONG?

PSSR missing prior to PEM.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 30 Jul 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Skimmer tank overflow remains contained in the tank enclosure.

WHAT WENT WRONG?

- Incident caused by unscheduled power outage due to electrical failure at Transformer Station
- Inadequate infrastructure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 03 Sep 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Breaking Containment Locations: Loading/unloading coupling
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During manoeuvres for the start-up, working on the different lines that contribute to the 32" collector, a sudden increase in pressure was observed in the pumping system, which generated a rupture of the filter O-ring corresponding to the UAM and subsequently a pumping stoppage was evident towards the Pumping Header.

WHAT WENT WRONG?

Equipment failure/inadequate protection.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Not available.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 26 Sep 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We recognize change.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Breakage of the truck's loading hose when closing the tank valve while the installation loading valve was open with the pump on.

WHAT WENT WRONG?

Inadequate communication between work teams

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 07 Sep 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Loss of production due to 1/2-inch valve in pump discharge.

WHAT WENT WRONG?

Improper adjustment/repair/maintenance.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 17 Oct 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We recognize change.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Loss of oil pipeline in lug branch, in transition from steel to fiber.

WHAT WENT WRONG?

Inadequate change management

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

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DATE: 26 Oct 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Unspecified.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Loss in pump discharge piping

WHAT WENT WRONG?

Work planning

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 06 Dec 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil loss in pipeline that injects chemical product into bba drive.

WHAT WENT WRONG?

Improper design development.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 14 Nov 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Flare and atmospheric vent systems (intended discharge location)
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

It vents through the flare and drags liquid in the form of a spray.

WHAT WENT WRONG?

Control system failure

CORRECTIVE ACTIONS & RECOMMENDATIONS:

-

BARRIERS:

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 14 Apr 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We stay within operating limits.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During production start-up of Battery, two hours after the start of the manoeuvres, a gas leak was detected through the liquid level control valve of the Slug Catcher due to erosion.

WHAT WENT WRONG?

- Lack of sand removal equipment.
- Well opening speed facilitated the removal of accumulated solids.
- Sand Trap design, purging of solids and drainage of liquid not suitable for real conditions.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Sand removal strategy review.
- Review of Sand Trap design, solids purge and battery level control valve.
- Gas collection pipelines cleaning.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 30 Dec 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Operators were performing manual draining of bottom liquid from the glycol tower when a check valve on the 1-inch drain line broke and a gas leak occurred. The lines were subsequently depressurized and the glycol tower went out of service.

WHAT WENT WRONG?

- Glycol tower not designed for solid (sand) presence.
- Due to solids, drain system was manually used instead of automatic level control.
- Delay in stopping the leak due to non-use of emergency buttons.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Improve drain system and level control design.
- Write SOP for manual level control.
- Glycol tower cleaning.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 14 Dec 2023
COUNTRY: Argentina
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Flexible hose/piping
PROCESS SAFETY FUNDAMENTAL: We recognize change.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At approximately 04:00, personnel from the loading area were carrying out routine tasks. During their rounds, they noticed that a pump was turned off and proceeded to inspect it. At that moment, they observed a condensate leak due to the detachment of a 4-inch hose, which was channelling product from the booster pump to the turned-off pump.

WHAT WENT WRONG?

It is a temporary installation that had little design/construction time (due to Business Unit's context).

A rigid discharge pipe with a flanged coupling was planned. They did not arrive with that pipe and that coupling and implemented a spindle with a quick coupling, without proper management of change.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

The need for adequate management of changes between what was designed and what was built, even more so in temporary installations and with accelerated projects.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 30 Jan 2023
COUNTRY: Argentina
FUNCTION: Unspecified
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Other production
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Spill at a decommissioned 8 inch #150 line, isolated from the process through a blind flange that failed due to internal corrosion.

During manual excavations to inspect the decommissioned 8 inch pipeline to find an oil release source, the workers removed the external coating and the fluid was released through a hole of 5 cm diameter due to internal corrosion. The fluid was 98% crude oil, 2% water. The estimated total oil volume spilled was 82 barrels, affecting 135 square metres of land surface.

WHAT WENT WRONG?

Management of Change not utilized, Potential consequences not understood.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Always use the "Pipeline Abandonment and Decommissioning Standard" to put a line (internal line or pipeline) out of service and manage through MOC.

Always conduct Work Planning and Risk Assessment for source release inspection work orders.

For future facilities minimize underground process lines in order to eliminate external corrosion risk.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 18 May 2023
COUNTRY: Bolivia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During an unauthorized gas venting manoeuvre in a well pad, a flammable atmosphere was generated which found an ignition source and a deflagration occurred which affected a worker.

WHAT WENT WRONG?

Lack of controls for release of gas from the well directly into the atmosphere, which is a deviation from company procedures.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Strengthen safety culture,
- Improve deficiencies in training and education, lack of clear expectations, work overload, and poor communication.
- Encourage contractors to conduct learning workshops.

BARRIERS:

Hardware Barrier Failures: Ignition Control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 21 Apr 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore flowline
PROCESS SAFETY FUNDAMENTAL: We recognize change.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil leak due to a hole in the section of the production flow line from the production wells to the oil gathering station.

WHAT WENT WRONG?

- Inadequate identification of necessary actions for risk mitigation and prevention.
- Lack of identification of the need for the development of an inspection plan.
- Line was not covered by an inspection plan. The project did not foresee the possibility of low fluid flow, as there are other lines that connect the well production to the collection station. As there are periods of low flow, water is formed by the stagnant oil separation process, accelerating the corrosion process.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Review the documentation of piping systems, examining flowcharts and the line classification list.
- Provide training for personnel involved in management of change.
- For pipeline sections susceptible to corrosion due to stagnant or low-flow conditions, propose a new inspection methodology for detecting areas with localized thickness loss.
- Review hazard analysis of separation vessel facilities containing looped pipelines to anticipate no flow scenarios and their consequences.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 14 Jun 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Leakage of 10 cubic metres of crude oil at junction of 4-inch pipeline from Station A with 12-inch pipeline from Station B, reaching an area of approximately 100 cubic metres in permeable soil.

WHAT WENT WRONG?

- Corrosion on nipple thread due to a deficiency in surface finish, as there were micro defects inserted in the manufacturing process.
- Pitting corrosion caused by absence of a chemical barrier (protection) in the assembly without use of grease on the thread of installed nipple.
- Design errors: there is no interlock and emergency shutdown to actuate protection instrument (pressure switch) connected to the system, the component specification (plug nipple) is not adequate, and line supports need to be better evaluated in terms of constant vibration in the location.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Install new pipe taps with standardized tubes, carry out a non-destructive testing on threads and issue a tube integrity report.
- Interlock PSH and PSL instruments with station transfer pumps.
- Change the location of corrosion coupons holders in pipeline/station junction and improve supports of pipes where they will be installed.
- Carry out vibration analysis at 1-inch corrosion monitoring points of pumping system operated by alternative pumps (similar to the event) to verify the need for improvements in the support of the entire operating unit.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Shutdown Systems – including operational well isolation and drilling well control equipment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 21 Feb 2023
COUNTRY: Colombia
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Pipeline operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Onshore Pipelines/Flowlines: Onshore pipeline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

At 08:24 on 21 February 2023, a pressure drop in the pipeline was indicated in the control room.

At 08:35 the owner of one of the lands reported that a backhoe working on his land had struck the pipeline. The pipeline was ruptured resulting in a loss of containment of 665 bbl of crude oil.

Primary containment of the spill was in the artificial channel created by the machinery that ruptured the pipeline.

It should be noted the barriers in place are what would be expected for managing a right of way and this event was caused by negligence by the landowner when instructing the operator of the backhoe to excavate the drainage channel.

WHAT WENT WRONG?

Hazard not identified by the landowner when instructing the contractor to excavate the drainage channel.

Lack of / poor communication the operator of the backhoe had not been made aware of the right of way.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

1. Ensure that further training is carried out with the landowners to ensure that they properly prepare any excavation work near to the right-of-way.
2. Ensure that the frequency of inspections is sufficient to identify any excavation work that may be planned. Consider increasing inspection frequency in periods of the year when landowners are more likely to be performing maintenance work.
3. The format of the security report could be modified to highlight any issues and make the report easier to digest.
4. Patrol reports by Security have been reviewed in which excavations have been reported in the right-of-way between the surveyed kilometres posts. However, there is no evidence that there is an action protocol that is activated in response to these reports. In this way, it is possible to immediately go to the farm, verify that it is an illegal excavation and talk to the owner in a timely manner.
5. Maintain an updated registry of farm owners. Maintain permanent contact with farm owners from community relations to detect possible changes in ownership. Focus awareness campaigns on the danger of illegal activities in the right-of-way.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

DATE: 01 Jul 2023
COUNTRY: Peru
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the maintenance handover operation of a valve, the manual shut-off was closed, resulting in a gas leak from one of the vents. The evacuation alarm was triggered, and the line was depressurized, leading to the shutdown of the plants.

WHAT WENT WRONG?

Maintenance management, inspection, reliability: incorrect inspection process or cycle. The execution of none of its stages (pre-commissioning, commissioning, PSSR).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

During plant shutdowns, make sure to correctly execute the inspection and testing of ball valves according to established procedures.

BARRIERS:

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 27 Dec 2023
COUNTRY: Brazil
FUNCTION: Drilling and Completion Operations
NUMBER OF DEATHS: 0
ACTIVITY: Workover/Well services
MODE OF OPERATION: Abandonment
POINT OF RELEASE: Wells, drilling and intervention: Well
PROCESS SAFETY FUNDAMENTAL: We recognize change.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During removal operation of the COP (production column), after disconnection of production tube with nozzle and 3.5-inch plug body installed, observed loss of containment of 6.36 cubic metres of oil that was confined in the COP, below the plug, due to imbalance. From the total volume, 6.3572 cubic metres was contained in the drilling floor and 0.0028 cubic metres was discharged into the sea.

WHAT WENT WRONG?

- Failure to identify change (MOC).
- Failure to perform the project risk analysis.
- Well construction failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Event still under investigation, with no recommendations developed so far.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 11 Jan 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Relief, Vent and Discharge Systems: Drain
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

LOPC of flammable gas, gas condensate and glycol occurred when opening the drain of three-phase gravity separator level transmitter. There was a flow of cold gas directed to the employee's forearm, who was taken to hospital for evaluation.

WHAT WENT WRONG?

- Risk analysis recommendation not implemented – no hose was installed to direct the drained flow to closed drainage system.
- Operating procedure with gaps – level gauge drainage activity is not detailed in the procedure for monitoring and controlling the vessel interface level.
- The risks and impacts involved in the activity are not listed – task risk assessment was not performed.
- Lack of valve inspection and preventive maintenance plan – instrument drain valves were stuck, making opening and closing difficult.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Re-evaluate Risk Analysis and, if necessary, propose new recommendations.
- Review and disclosure of procedure including risks and impacts involved in the activity.
- Carry out corrective maintenance and implement systematic inspection and maintenance of drain and block valves.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Overexertion or improper position/posture for task

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 13 Mar 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During normal operation of the gas turbine power generation, gas detectors located in ventilation duct of the turbine hood were activated, indicating presence of inflammable gas.

WHAT WENT WRONG?

After vibration tests and analysis, the vibration sources identified are acoustic pulsations in combustion chamber related to CO_2 concentration connected to high flow/high pressure inside the system. The presence of loose brackets is a contributor to increasing the tubes vibration.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Install an orifice on secondary purging line. Orifice definition by checking vibration and triaxial accelerometers installed on J-tuber/manifold running with current fuel gas composition up to 30% CO_2. A dedicated study and test campaign shall be conducted to confirm the presence of acoustic pulsation in combustion chamber and address any potentially needed design change. Forward analysis for future projects to implement a vibration reduction solution in turbines.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

DATE: 15 Mar 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During normal operation of the gas turbine power generation, gas detectors located in ventilation duct of the turbine hood were activated, indicating presence of inflammable gas. The event is the same as occurred on 13 March 2023.

WHAT WENT WRONG?

After vibration tests and analysis, the vibration sources identified are acoustic pulsations in combustion chamber related to CO_2 concentration connected to high flow/high pressure inside the system. The presence of loose brackets is a contributor to increasing the tubes vibration..

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Install an orifice on secondary purging line. Orifice definition by checking vibration and triaxial accelerometers installed on J-tuber/manifold running with current fuel gas composition up to 30% CO_2. A dedicated study and test campaign shall be conducted to confirm the presence of acoustic pulsation in combustion chamber and address any potentially needed design change .Forward analysis for future projects to implement a vibration reduction solution in turbines.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 09 Apr 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During normal operation of the gas turbine power generation, gas detectors located in ventilation duct of the turbine hood were activated, indicating presence of inflammable gas. The event is the same as occurred on 13 March 2023 and 15 March 2023.

WHAT WENT WRONG?

After vibration tests and analysis, the vibration sources identified are acoustic pulsations in combustion chamber related to CO_2 concentration connected to high flow/high pressure inside the system. The presence of loose brackets is a contributor to increasing the tubes vibration.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Install an orifice on secondary purging line. Orifice definition by checking vibration and triaxial accelerometers installed on J-tuber/manifold running with current fuel gas composition up to 30% CO_2. A dedicated study and test campaign shall be conducted to confirm the presence of acoustic pulsation in combustion chamber and address any potentially needed design change. Forward analysis for future projects to implement a vibration reduction solution in turbines.

BARRIERS:

Hardware Barrier Failures: Process Containment

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Monitoring, reporting and learning

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 08 May 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Oil spill detected near the poop deck, at the stern of the vessel. The source of the leak was identified through a diving operation. Flow interrupted by sea chest plugging. Oily residue reached the coast.

WHAT WENT WRONG?

There is no operating procedure with a pump for flushing. Failure to carry out MOC to evaluate the use of a pump. Absence of check valve in pump discharge line to prevent contaminated liquid from returning to the sea chest.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Develop a procedure for operating this make of pump.
- Technically evaluate the installation of a check valve in the pump discharge line, after its bypass line.
- Install check valves in the lines of systems originating from sea chest and/or other interconnected systems.
- Check the integrity of the sea chest and interconnecting valves, as well as carry out a critical analysis of the effectiveness of valves maintenance plans.
- Evaluate emergency response actions for offshore events related to the release of oil at sea.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Emergency Response Equipment and Systems

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Human Barrier Failures: Authorization of temporary and mobile equipment

Human Barrier Failures: Response to emergencies

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 16 Jun 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping joint
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Loss of containment of 3.3 cubic metres of ethanol in flange of the suction line of an ethanol pump. The pump was stopped immediately and the entire leaked volume was contained in skid and directed to the unit's drainage system. There was a failure in actuation of the check valve in the pump suction line.

WHAT WENT WRONG?

- Loose flange on pump suction pipe. It was identified that the check-valves broke due to the vibration that naturally occurs in the equipment.
- The breakage of the check-valves led to an increase in vibration and consequently to a loosening of the flange bolts in the pump suction.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Perform a complete vibration analysis on the chemical and sea water systems to identify where more supports are needed to reduce vibration. Disclose the result of the vibration analysis to other units/projects.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: 01 Jul 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the commissioning of a gas compression plant, tubing detached from the pressure instrument in discharge of the main compression system. This gas leak was detected by the fire and gas system, which caused an emergency shutdown of the process plant.

WHAT WENT WRONG?

Leakage through PIT process hook-up; connection not mapped or verified or tested after Leak Test (it was a common understanding that the affected tie-in point was not part of the witness joint list and had no inspection or verification prior pressurizing the system with HC); lack of training or tools to assure that the connection has been properly made; assembly failure of the Tubing/OD connector in the process 2-OD connector with incorrect assembly during Construction Phase.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Comprehensive Internal Inspection in all tubing OD connections that have been used as vent or pressurization points in the leak test regarding HC or High-Risk Inventory (not in operation). This recommendation is impeditive for start-up (HC pressurization) of systems under commissioning.
- Guarantee that all connections used during test phase regarding HC or High-Risk Inventory are reliable and not leaking. Qualified Personnel shall be used for this activity, using tools to identify connection problems and possible leakage points.
- Update the witness joint list including all points that have not yet been pressurized and not yet commissioned. This recommendation is impeditive for start-up (HC pressurization) of systems under commissioning.
- Revise the Inspection Template to include inspection details or the use of Inspection Gauge after the assembly. Current Template only indicates inspection and refers to the Project Hook-Up. It is necessary to include a more detailed checklist item requiring proper inspection of the connection. It must be evaluated if it is valid to also include Photo Requirement using the Gauge for inspecting.
- Assure training for Instrumentation Technicians for Tubing Assembly during Construction Phase.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 06 Jul 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Start-up
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

During leak test of oil treater hydrocyclones, a flanged connection was opened improperly and there was a LOPC of 100 kg of heated produced water. Part of this water hit the operator who was performing the manoeuvre and caused a second-degree burn to his right calf.

WHAT WENT WRONG?

- One of the hydrocyclones' oil outlet lines had not been previously tested. This line, which was not tested initially, had a weak connection in the hose that connects the waste oil to the hydrocyclone.
- All valve alignments of lines arriving at hydrocyclone were not checked to carry out all leak tests.
- Process fluid (hot water) was used to perform the leak test.
- There is no procedure or standard that describes how hydrocyclone system should be cleaned and returned to operation.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Create a procedure or standard that describes maintenance and recommissioning steps of the hydrocyclone system.
- Train operation staff and contractors who need to perform the task.
- Do not use process fluids to perform leak tests.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

NONE: None: Unspecified

DATE: 16 Jul 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During plant normal operation, emergency shutdown was activated due to confirmed gas detection on forward main deck area. P/V breaker water seal released, allowing gas from the cargo system tanks to scape to atmosphere.

WHAT WENT WRONG?

- Unawareness of low level or high level or ideal level on PV breaker water seal, lack of monitoring.
- Procedures and daily routine visual inspections were not followed, lack of robust equipment and accessories, lack of equipment design details, instructions and specific manual.
- Inadequate communication, inadequate leadership/supervision, inadequate maintenance, inadequate work standard, inadequate robust daily visual checks, lack of robust periodical monitoring.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Marine Superintendent to provide on board training of the marine team on the methods for verification/check of PV breaker during normal operations and the importance of the safety rounds on site.
- Creation of a Work Instruction for PV breaker monitoring based on the PV Breaker Manual instructions.
- Check with Chemical department on what type of liquid should be used in the PV Breaker seal to provide optimum and correct efficiency based on the density. Such as but not limited to fresh water, salt water, mix of liquid with 30% ethylene glycol or any other to be evaluated and recommended.
- Perform integrity assessment on PV Breaker check previous reports and condition of temporary repair.

BARRIERS:

Hardware Barrier Failures: Protection Systems – including deluge and fire water systems

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

DATE: 24 Jul 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Valve (body, stem, plugs)
PROCESS SAFETY FUNDAMENTAL: Not applicable.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

An operator was in the injection seawater treatment module awaiting instructions to perform operational manoeuvres in the area, when he was hit by a jet of desulphated injection water coming from a hole in a valve body of downstream seawater removal unit sulphate.

The operator had to be temporarily removed from work for a few days due to the injury he suffered.

WHAT WENT WRONG?

Specification of valve with material incompatible with the fluid, requiring use of internal casing.

The project did not consider reducing human exposure to consequences of possible equipment or system failures – lack of automation in valves located in sulphate removal unit, which requires manoeuvres to be carried out on site.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Replace valves with organic casing, located in the sulphate removal unit system. Review project specifications to abolish use of organic internal casing in valves.

Carry out inspection (Ultrasound or X-ray) to monitor the thickness reduction in valves bodies with organic coating – technical authority area will be responsible for defining the best method.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 21 Aug 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the start-up of the plant, there was detachment of tubing from the flow meter in the gas injection module. This gas leak was detected by the fire and gas system, which caused an emergency shutdown of the FPSO.

WHAT WENT WRONG?

Lack of verification of tubing assembly in systems with HC or high-pressure inventories. Lack of training or tools to assure that the connection has been properly made; assembly failure of the Tubing/OD connector in the process.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Comprehensive Internal Inspection in all tubing OD connections that have been used as vent or pressurization points in the leak test regarding HC or High-Risk Inventory (not in operation). Provide tools for all types of tubing. Provide training for the applicable positions on the correct assembly of tubing and connections and include in the competency matrix, as well as include a more detailed checklist item requiring proper inspection of the connection.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Human Barrier Failures: Acceptance of handover or restart of facilities or equipment

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 01 Aug 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Emergency shutdown
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

An employee, contractor or subcontractor 'days away from work' injury and/or fatality

INCIDENT DESCRIPTION:

During execution of an installation of brackets on tubing assembly, there was a loss of gas containment, after a tubing detached from the gas export system. Three employees who were working in the area suffered injuries.

WHAT WENT WRONG?

- Inadequate certification of materials.
- Absence of minimum qualification and/or assembly and testing tubing training programme.
- Inadequate qualitative or quantitative risk identification and analysis (recurrence).

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Replace the inadequate connections that were installed.
- Lock and isolate low-quality connections in the warehouses of the Contractors.
- Request a list of all services performed that may have involved the installation of new connections on the operating platforms, with identification and verification of manufacturers.
- Conduct risk analyses for decoupling risk scenarios.
- Training and dissemination for the entire workforce.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Execution of activities

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 07 Sep 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Unspecified – other
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Instrumentation and small bore tubing
PROCESS SAFETY FUNDAMENTAL: We apply procedures.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During normal operation on the platform, there was a loss of gas containment by the sealing system of the reinjection compressor resulting in emergency shutdown. The process plant was depressurized, and the platform's emergency alarm was triggered.

WHAT WENT WRONG?

- Project information not properly recorded.
- Failure in verification of tubing assembly in systems with HC or high-pressure inventories.
- Failure in training or tools to ensure a correct assembly of these connections.
- The mechanical lock (interlock) was not correctly mounted.
- Improper operating procedure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Comprehensive inspection (on all OD connections) in systems with HC or high-pressure inventories.
- Provide training and checklist for the applicable positions on the correct assembly of tubing and connections.
- Attest, with the package provider, that the logics linked to this scenario are functional.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Organization, resources and capability

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 11 Sep 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During replacement of dresser joint sealing of offloading header, there was a leak of approximately 7.5 cubic metres of residual oil from the line.

WHAT WENT WRONG?

The drain obstruction check was not carried out and there is no procedure for such activity as required by the equipment release standard. Lack of cleaning procedure for opening the offloading line. It was identified that the team involved in the WP did not expect the line to contain remaining oil. Isolation plan failure.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Develop and train operators in the procedure of cleaning, draining and releasing the Offloading discharge lines that require opening the system for maintenance.
- Perform a leak test of the Offloading header valves, and maintain those that fail.
- Perform isolation standard recycling and WP.
- Include in this training the use of the unit's risk analysis as an additional tool for analysis and completion of the risk matrix in WP.

BARRIERS:

Hardware Barrier Failures: Process Containment

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 26 Sep 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Tanks and Sumps/Pits: Atmospheric tank overflow
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

During the manoeuvre of transferring oily water between cargo tanks, there was an inadvertent increase in the level of the slop tank, due to two valves open at this moment, with consequent loss of primary containment of 2.8 cubic metres of oil on the deck.

WHAT WENT WRONG?

- Failure to constrain the tank level transmitter listed in the Critical Elements List.
- Improper operating procedure.
- Manufacturer's recommendation and standard for inspection, testing, and maintenance procedure not implemented.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Check performance of cargo tank level gauges and transmitters.
- Review procedure for fluid transfer manoeuvres between the unit's cargo tanks, including all possible manoeuvres between cargo tanks and slop tanks. Include in this procedure specific contingency actions for failure of level sensors in cargo tanks.
- Implement interlocking according to the logic presented by the automation.

BARRIERS:

Hardware Barrier Failures: Process Containment

Hardware Barrier Failures: Detection Systems

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 27 Dec 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Piping in Process and Utility Systems (excluding subsea) : Piping material/tubing
PROCESS SAFETY FUNDAMENTAL: We stop if the unexpected occurs.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Interruption of production plant due to identification of a gas leak in the inlet piping of the gas regeneration vessel inlet line.

WHAT WENT WRONG?

It was identified that the failure mode of the event was due to the material being carbon steel (inadequate design) combined with the change in interaction with the material due to changes in the fluid.

Additionally, a human cause was seen in the human decision to do a controlled shut down rather than ESD, which led to a Tier 1 event.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Replace the material used from carbon steel to stainless steel.

Create a decision-making process standard for plant shutdown in cases of hydrocarbon leaks.

Implement process engineering recommendations to address problems encountered regarding fluid-material interaction.

BARRIERS:

Hardware Barrier Failures: Structural Integrity

Human Barrier Failures: Response to process alarm and upset conditions (e.g. outside safe envelope)

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 25 Jan 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Meter
PROCESS SAFETY FUNDAMENTAL: We watch for weak signals.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

An alarm for mustering was triggered following several detectors that picked important gas concentration.

WHAT WENT WRONG?

- Wrong O-ring installed: O-ring sized for 100 bars instead of 690 bars.
- Spare parts management.
- Spare Parts SAP incomplete configuration.
- Not suitable HP Flowmeter Barrier design.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Recommendations:

1. Input good parametrization on SAP triggering the need to purchase the spare part.
2. Ensure that for planned operations there are available spare parts in case it is necessary to change any part.
3. Assess the feasibility of increasing BDV capability and associated piping and systems to ensure pressure can be reduced quicker.
4. Assess the possibility to change to another HP flowmeter design considering that FT-2430 is the only single chamber in the plant and only has one barrier to environment which is broken every time it is necessary to change the orifice plate or to inspect.

BARRIERS:

Human Barrier Failures: Operating in accordance with procedures – PTW, Isolation of equipment, Overrides and inhibits of safety systems, Shift handover, etc.

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 05 Aug 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Equipment: Pump
PROCESS SAFETY FUNDAMENTAL: We sustain barriers.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Pumpman revisited the pump room and observed the loss of primary containment originating from vacuum pump top cover gasket. Without delay, he promptly communicated the exact location of the LOPC to the Marine Control Room Operator (MCRO) who was on duty at the time. Recognizing the significance of the situation, the MCRO swiftly initiated a halt to the ongoing offloading operations.

WHAT WENT WRONG?

The root cause of the LOPC was identified as a passing valve that pressurized the vacuum pump system, leading to the gasket failure on the vacuum pump cover. The valve is not in the maintenance programme.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Disclose the event with the wellheads in the field in order to raise awareness.
- Replace gasket for the vacuum pump cover to rectify the issue.
- Perform backflush between the tanker and the offload terminal.

BARRIERS:

Human Barrier Failures: Surveillance, operator rounds and routine inspection

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 11 Sep 2023
COUNTRY: Brazil
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Normal
POINT OF RELEASE: Subsea: Subsea pipeline/flowline
PROCESS SAFETY FUNDAMENTAL: We respect hazards.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

The Control Room Operators observed a dropping trend on fuel gas import flowline. The OIM was contacted and aligned to changeover the GTG-A from gas to diesel, since the gas pressure was still dropping. After the GTG-A fuel changeover was completed, the platform decided to shut the topside import gas valve.

External damage to the Gas Import Pipeline (GIP) was confirmed. No one was injured and the gas leak was stopped.

WHAT WENT WRONG?

- Safety and security measures/marking/sign/warnings – inadequate protective or safety measures: The gas import pipeline was installed without protection.
- Undesired behaviour (internal and external) – careless/unfocused/not observant/distracted: External agent crossed the pipeline while dragging an anchor.
- Inadequate Risk Assessment prior to the activities: The risk analysis was not updated considering unintentionally dragged anchors.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

- Protection of pipelines, flowlines and other subsea structures that could be damaged.
- Include subsea structures maps and inform all fleet about the change.
- Prepare operation procedure for the control room operators to close the SSIV immediately on indications of a leak.
- Establish an ROV vessel strategy to secure short-notice mobilization.
- Identify and require update of governing documents, that needs to be updated.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 18 Feb 2023
COUNTRY: Guyana
FUNCTION: Production
NUMBER OF DEATHS: 0
ACTIVITY: Production operations
MODE OF OPERATION: Routine maintenance
POINT OF RELEASE: Breaking Containment Locations: Breaking containment location
PROCESS SAFETY FUNDAMENTAL: We maintain safe isolation.

CATEGORIES:

A release above threshold quantity in any 1 hour period

INCIDENT DESCRIPTION:

Methanol tank #1 was prepared and isolated for confined space entry from methanol tank #2. The purpose was to replace the level transmitter inside tank #1 sump. On the afternoon of 17 February, the first entry by the Rope Access Team was conducted to install a safety line for the instrument team's entry. Afternoon of 17 February, transmitter was identified as incorrect product. The confined space entry work was suspended on afternoon of the 17th and plans made to prepare the tank for service. While confined space entry suspended, troubleshooting on the hydraulic system occurred and the hydraulic return valve was closed instead of the supply valve. Valve configuration caused an increase in hydraulic pressure, actuating cross over valves allowing transfer from methanol tank #2 to tank #1. On 18 February, at approximately 16:00, the Control Room Operator (CRO) observed low readings on the GIRs at the Methanol Tank area and level changes on Methanol Tanks.

WHAT WENT WRONG?

- inadequate positive isolation completed for vessel entry: risk assessment did not address entry without positive isolation
- inadequate approval level for vessel entry without positive isolation
- incorrect valve isolation
- control lines not depressurized
- no indication to operator of valve position.

CORRECTIVE ACTIONS & RECOMMENDATIONS:

Update procedure to address positive isolation requirements including assessment process and proper approval levels. Clearly identify supply and return hydraulic valves.

BARRIERS:

Management System Element Barrier Failure: Risk assessment and control

Management System Element Barrier Failure: Asset design and integrity

Management System Element Barrier Failure: Plans and procedures

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

FATAL INCIDENTS CLASSIFIED AS PROCESS SAFETY EVENTS 2023

AFRICA ONSHORE

DATE: Jun 26 2023
COUNTRY: Nigeria
FUNCTION: Drilling
NUMBER OF DEATHS: 1
CAUSE: Struck by (not dropped object), ACTIVITY: Drilling, workover, well operations
RULE: Energy isolation
BODY PART: <<Body part not allocated>>
NATURE OF INJURY: <<Nature of injury not allocated>>

Employer: Contractor   Occupation: Manual labourer

NARRATIVE:

The activity on well 2 started on June 17th with the equipment mobilization in location. On June 18th the job officially commenced on site.

The Scope of Work was to carry out maintenance activities on Xmas Tree and ambient valve recalibration. Operation was performed with support of Slick Line service and downhole plug.

During the activity, the team observed the Slick Line and the upper sheave were not moving. In order to fix the issue, the Contractor Supervisor bled off the pressure in the hydraulic hose of the SL stuffing box through the dedicated hydraulic line, and proceeded to slack the cable at the bottom sheave to resume jar action This brought to failure the pressure equalization across the downhole plug and consequently led to the ejection of the same inside the lubricator which parted and jumped on air. The lubricator ejection determined the Slick Line cable tension and strong knockback. Consequently the lower sheave, connected to such cable, strongly impacted the IP's chest and chin. IP medical conditions immediately appeared critical. The event led also to a slight gas release.

WHAT WENT WRONG:

- Rigless Program developed PtW, Risk Assessment, Method of Statement, Isolation Certificate, TBT forms: all available in location, improvement needs have been observed in the procedures coordination and implementation.
- Lack of supervision.
- In presence of an issue (blocked Slick Line sheave), Contractor Supervisor decided to bleed off pressure in the hydraulic hose of the Slick Line stuffing box by pushing the small ball in the check valve seat allowing the hydraulic oil to bleed. This brought to the failure of the pressure equalization and following lubricator's ejection. Operator did not stop the activity to re-assess the Risks.
- The Contractor's personnel competences were not previously assessed before starting the activity due to lack of Procedures. During the contractor's personnel interviews carried out in the Investigation process competence's gaps appeared evident for some crews members.
- Lack of knowledge on First Aid Operations and Emergency Communications. Night MEDEVAC Operations to be improved.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- While dealing with high pressure fluids ensure proper barriers always in place.

- In case the assessed workflow (as per method of statement) changes, proper risk assessment must be done and mitigations identified. It must be treated as management of change.
- Whenever you are planning a job, always take in consideration the operational context and the process conditions.
- All operations (routinely and critical ones) require competent person to be performed; their knowledge must be formally assessed prior to start any job.
- In all locations medevac drills must be performed and personnel to be adequately informed and trained about the existent procedures to be implemented.
- Proper communication devices in remote areas are fundamental to manage any emergency situation.
- Ensure safe rescue by medevac of the workers during day & night.
- Stop work authority to be always applied whenever required (anyone is entitled).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

AFRICA OFFSHORE

No fatal incidents classified as process safety events.

ASIA / AUSTRALASIA ONSHORE

No fatal incidents classified as process safety events.

ASIA / AUSTRALASIA OFFSHORE

No fatal incidents classified as process safety events.

EUROPE ONSHORE

No fatal incidents classified as process safety events.

EUROPE OFFSHORE

No fatal incidents classified as process safety events.

MIDDLE EAST ONSHORE

DATE: Jun 28 2023
COUNTRY: Kuwait
FUNCTION: Unspecified
NUMBER OF DEATHS: 1
CAUSE: Pressure release, ACTIVITY: Production operations
RULE: Line of fire
BODY PART: <<Body part not allocated>>
NATURE OF INJURY: <<Nature of injury not allocated>>

Employer: Contractor   Occupation: Process/equipment operator

NARRATIVE:

When a Production crew was conducting routine well monitoring and troubleshooting, they noticed a spill from a nearby pipeline. The crew approached the spill site to investigate. Upon arriving at the site, the crew saw the water disposal pipeline had failed, and a pinhole was sending high-pressure produced water into the desert sand, creating a five metre-wide and two metre-deep whirlpool. During this observation, one of the workers slipped into the newly formed whirlpool. Crew members made several attempts to rescue him and were finally able to retrieve him from the water. One of the rescuers went into the hole and had to be rescued without any further harm. Emergency Medical arrived on site to begin advanced life-saving techniques but were unsuccessful.

WHAT WENT WRONG:

Unaware of surroundings.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

Process safety fundamental: We respect hazards

<<No Causal Factors Allocated>>

DATE: May 2 2023
COUNTRY: Oman
FUNCTION: Unspecified
NUMBER OF DEATHS: 1
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature, ACTIVITY: Drilling, workover, well operations
RULE: Other issue – no applicable rule
BODY PART: <<Body part not allocated>>
NATURE OF INJURY: <<Nature of injury not allocated>>

Employer: Contractor   Occupation: Manual labourer

NARRATIVE:

At approximately 2:55 am, a contractor was performing circulation prior to cement on well, when the wellhead started leaking fluid with hydrogen sulphide (H2S) release. A total of 7 people were reported to be impacted (1 deceased, 4 transferred to hospital and stable, 2 examined at site clinic). The well was secured (BOP shut-in) and the location was evacuated.

WHAT WENT WRONG:

Death from H2S exposure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Unspecified.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

MIDDLE EAST OFFSHORE

No fatal incidents classified as process safety events.

NORTH AMERICA ONSHORE

DATE: Feb 24 2023
COUNTRY: USA
FUNCTION: Production
NUMBER OF DEATHS: 1
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature, ACTIVITY: Maintenance, inspection, testing
RULE: Bypassing safety controls
BODY PART: <<Body part not allocated>>
NATURE OF INJURY: <<Nature of injury not allocated>>

Employer: Company Occupation: Process/equipment operator

NARRATIVE:

Worker was managing high fluid levels in the low pressure knock out (LPKO) vessel while bringing wells online at the facility and was found unresponsive.

WHAT WENT WRONG:

Unexpected release.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Review small building design with focus on ventilation and alarms. Reinforce existing practice of not draining process fluids within enclosed spaces or near ignition sources. Reinforce use of personal multi-gas protection monitors to verify acceptable atmospheric conditions, especially prior to building / enclosure entry and during activities involving process fluids. Ensure lone worker policies and safety-critical tools are functioning as intended –

including calibration and testing – and supervisors are accountable for monitoring use and adherence.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

NORTH AMERICA OFFSHORE

No fatal incidents classified as process safety events.

RUSSIA & CENTRAL ASIA ONSHORE

No fatal incidents classified as process safety events.

RUSSIA & CENTRAL ASIA OFFSHORE

No fatal incidents classified as process safety events.

SOUTH & CENTRAL AMERICA ONSHORE

No fatal incidents classified as process safety events.

SOUTH & CENTRAL AMERICA OFFSHORE

No fatal incidents classified as process safety events.

FATAL INCIDENTS RELATED TO PROCESS SAFETY BUT NOT CLASSIFIED AS PROCESS SAFETY EVENTS 2023

AFRICA ONSHORE

No fatal incidents related to process safety but not classified as process safety events.

AFRICA OFFSHORE

No fatal incidents related to process safety but not classified as process safety events.

ASIA / AUSTRALASIA ONSHORE

DATE: Jun 22 2023
COUNTRY: Pakistan
FUNCTION: Production
NUMBER OF DEATHS: 2
CAUSE: Explosion, fire or burns, ACTIVITY: Transport – Land
RULE: Driving
BODY PART: <<Body part not allocated>>
NATURE OF INJURY: <<Nature of injury not allocated>>

Employer: Contractor	Occupation: Transportation operator
Employer: Contractor	Occupation: Transportation operator

NARRATIVE:

Contractor bowser, filled with crude oil, left the plant around 3 pm. While commuting from the plant to a refinery, at 10.45 pm, the bowser rolled over and caught fire, resulting in the driver's death. The helper suffered a burn injury and was immediately transported to hospital but later also lost his life.

WHAT WENT WRONG:

System of supervision was ineffective.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Introduce the continuous speed check for HAZMAT transportation.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

ASIA / AUSTRALASIA OFFSHORE

No fatal incidents related to process safety but not classified as process safety events.

EUROPE ONSHORE

No fatal incidents related to process safety but not classified as process safety events.

EUROPE OFFSHORE

No fatal incidents related to process safety but not classified as process safety events.

MIDDLE EAST ONSHORE

No fatal incidents related to process safety but not classified as process safety events.

MIDDLE EAST OFFSHORE

No fatal incidents related to process safety but not classified as process safety events.

NORTH AMERICA ONSHORE

No fatal incidents related to process safety but not classified as process safety events.

NORTH AMERICA OFSHORE

No fatal incidents related to process safety but not classified as process safety events.

RUSSIA & CENTRAL ASIA ONSHORE

No fatal incidents related to process safety but not classified as process safety events.

RUSSIA & CENTRAL ASIA OFFSHORE

No fatal incidents related to process safety but not classified as process safety events.

SOUTH & CENTRAL AMERICA ONSHORE

No fatal incidents related to process safety but not classified as process safety events.

SOUTH & CENTRAL AMERICA OFFSHORE

No fatal incidents related to process safety but not classified as process safety events.

HIGH POTENTIAL EVENTS RELATED TO PROCESS SAFETY BUT NOT CLASSIFIED AS PROCESS SAFETY EVENTS 2023

No high potential events related to process safety but not classified as process safety events reported.

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International Association of Oil & Gas Producers

The Process Safety Event data presented in this report are based on voluntary submissions from participating IOGP Member Companies and are not necessarily representative of the entire upstream oil and gas industry.

The Process Safety Events (PSE) data presented are based on the numbers of Tier 1 and Tier 2 PSE reported by participating IOGP Member companies, and are categorised by:

- onshore and offshore
- drilling and production
- activities
- consequences
- material released

https://data.iogp.org

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